

Muown: Row-Norm Control for Muon Optimization

Kai Lion¹ Florian Hübler^{1,2} Bingcong Li¹
Antonio Orvieto³ Niao He¹

¹ Department of Computer Science, ETH Zurich, Switzerland

² Department of Mathematics, School of Computation, Information and Technology;
Technical University of Munich, Germany

³ ELLIS Institute Tübingen, MPI-IS, Tübingen AI Center, Germany

{kai.lion,florian.huebler,bingcong.li,niao.he}@inf.ethz.ch
antonio@tue.ellis.eu

ABSTRACT

Muon has emerged as a strong competitor to AdamW for language model pre-training, yet its behavior at scale is sensitive to weight decay. Recent work has observed that, for Muon without decoupled weight decay, the spectral norm of weight matrices drifts upward over training. Through a decomposition of the spectral norm into a row-magnitude factor and a row-coherence factor, we identify the former as the empirical driver of this drift under Muon, while the latter remains well-behaved along the trajectory. Motivated by this diagnosis, we introduce Muown, a drop-in replacement for Muon that treats the row-magnitude vector as an explicit optimizer variable, updating it under the ℓ_∞ geometry induced by the decomposition, while applying Muon unchanged to the remaining direction component. We prove that Muown attains the optimal non-convex rates in both deterministic and stochastic regimes under a dual norm aligned with the underlying geometries and with a stochastic noise coefficient that empirically remains below that of Muon throughout training. Across GPT-style pre-training on FineWeb-Edu with model sizes from 124M up to 2.7B parameters, Muown improves perplexity over Muon, SOAP, AdamW, and Lion. It also widens the plateau of near-optimal learning rates across model scales, reduces sensitivity to weight decay, and avoids the spectral norm drift at negligible step-time overhead when appropriately sharded.

1 INTRODUCTION

Moving beyond the diagonal conditioning inherent to Adam and its variants (Kingma and Ba, 2015; Loshchilov and Hutter, 2019), an emerging class of optimizers is engineered to respect the matrix-shaped structure of the linear layers of the Transformer architecture. Notable examples include Shampoo (Gupta et al., 2018), SOAP (Vyas et al., 2025), Scion (Pethick et al., 2025) and Muon (Jordan et al., 2024b). In particular, Muon has been motivated through the perspective of steepest spectral descent (Carlson et al., 2015), which computes the update direction as the solution to the optimization problem $\arg \min_{\Delta \mathbf{W} \in \mathbb{R}^{m \times n}} \langle \mathbf{G}, \Delta \mathbf{W} \rangle$ s.t. $\|\Delta \mathbf{W}\|_{S_\infty} \leq 1$ where $\mathbf{G}$ is the matrix-shaped gradient of the loss with respect to $\mathbf{W} \in \mathbb{R}^{m \times n}$ and $\|\cdot\|_{S_\infty}$ is the spectral (Schatten- ∞) norm.

Despite the encouraging empirical success of Muon, it has been observed that, at scale, its formulation can suffer from numerical overflows in bfloat16 (Liu et al., 2025) and exploding attention logits (Kimi-Team et al., 2025). At the same time, the spectral norm of weight matrices has been documented to grow consistently throughout training under Muon without weight decay (Pethick et al., 2025, Fig. 9), violating the *spectral condition* for stable feature learning (Yang et al., 2024b), which prescribes $\|\mathbf{W}\|_{S_\infty} = \Theta(\sqrt{m/n})$ and $\|\Delta \mathbf{W}\|_{S_\infty} = \Theta(\sqrt{m/n})$. This drift away from the stable feature-learning regime could potentially be linked to the pathologies mentioned above.

A standard remedy is decoupled weight decay (Chen et al., 2025; 2024; Pethick et al., 2025), which keeps $\|\mathbf{W}_t\|_{S_\infty}$ in check by shrinking the weight matrix uniformly at every step. This intervention is non-targeted, scaling the entire spectrum uniformly, irrespective of which directions are

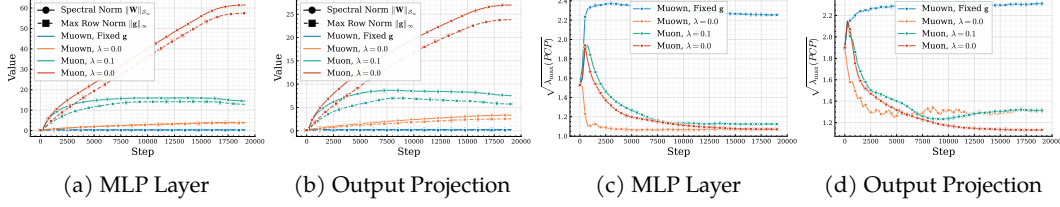


Figure 1: *Left two*: Drift of the maximum row norm $\|g_t\|_\infty$ drives the growth of the spectral norm $\|W_t\|_{S_\infty}$ for an MLP and attention output projection layer in a 500M transformer. When intervening by changing the parameterization s.t. the row magnitudes are fixed to the value at initialization, the systematic spectral norm increase vanishes. *Right two*: The remaining coherence factor $\lambda_{\max}(P_t C_t P_t)$ in the spectral norm exhibits no drift. λ denotes the weight decay, for more details, see Appendix A.4

responsible for spectral-norm growth, and has been observed to slow convergence in the early phase of training (Liu et al., 2025, Fig. 2), (Pethick et al., 2025, Fig. 8). We introduce a complementary perspective: rather than constraining the spectrum as a whole, we identify systematic row-norm growth as the empirical mechanism driving the spectral-norm drift (Fig. 1a, 1b) and control it by reparameterizing the weight so that the row scales become explicitly trainable. To remain agnostic to the model architecture, this reparameterization is realized implicitly inside the optimizer, leaving the forward pass unchanged. The resulting procedure achieves consistent gains in perplexity across all tested model scales and learning rates, at a minimal step-time overhead compared to Muon.

1.1 CONTRIBUTION

- We decompose the spectral norm into a row-magnitude factor and a row-coherence factor (Proposition 1), which singles out the row magnitude as the component whose evolution tracks the spectral norm under Muon. Building on this perspective, we propose Muown, a drop-in replacement for Muon that promotes the row magnitudes to an explicit optimizer variable, while Muon is applied unchanged to the directional component.
- Muown retains the optimal non-convex rates of Muon in both deterministic and stochastic settings (Theorems 1, 2), but established in the mixed dual norm matching the two update geometries. The result supports the empirical improvements of Muown through an improved noise term.
- On FineWeb-Edu pre-training from 124M to 2.7B parameters, Muown consistently improves perplexity over Muon, SOAP, AdamW, and Lion, maintaining a consistent perplexity advantage over the best-tuned Muon baseline. Further, it broadens the plateau of near-optimal learning rates across model widths (Fig. 5), and alleviates the need for weight-decay tuning (Fig. 4b). The method adds only minor step-time overhead once sharded (Table 3), matching Muon’s runtime in practical scenarios.

The remainder of this work is structured as follows. In Section 2, we review relevant background. We then introduce Muown in Section 3 and motivate it from the perspective of approximate spectral norm control via trainable row-scales. In Section 4, we establish convergence of Muown and conclude by comparing its effectiveness against Muon, SOAP, and AdamW in Section 5.

2 RELATED WORK

Shaping Gradient Spectra. Pre-conditioning is a central tool in optimization for machine learning (Bottou et al., 2018): while early approaches focus on efficient use of the Hessian or the Fisher Information (Martens and Grosse, 2015), contemporary methods directly operate on gradients, taking their matrix structure into account. An early notable work in gradient-based structure-aware optimizers is Shampoo (Gupta et al., 2018). Rather than applying pre-conditioning on flattened, structureless parameter vectors, Shampoo employs two matrix-shaped pre-conditioners. An idealized variant of Shampoo can be cast as performing steepest spectral descent under the spectral norm, effectively setting the singular values of the parameter update matrix to unity (Bern-

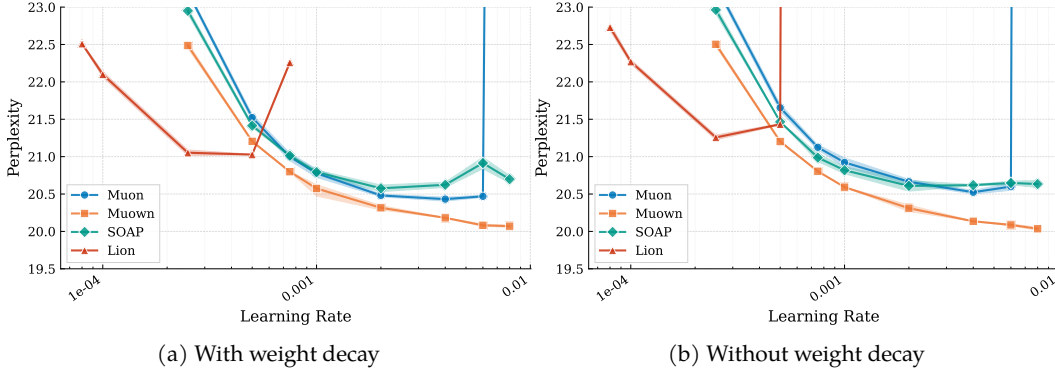


Figure 3: Perplexity as a function of learning rate for Muon, Muown, Lion, and SOAP on a 124M model. Muown achieves lower perplexity across all learning rates and is more robust to the choice of learning rate, maintaining strong performance even at higher values where Muon diverges. We repeat the average of two seeds, reporting min and max as shaded area. Details are reported in Appendix C.

stein and Newhouse, 2024). SOAP (Vyas et al., 2025) optimizes singular values adaptively with Adam. Most prominently, Muon (Jordan et al., 2024b; Tuddenham et al., 2022) shapes the spectrum through orthogonalization of the momentum state, approximating the normalized steepest descent direction (Boyd and Vandenberghe, 2004) under a spectral norm constraint (Carlson et al., 2015).

Operator Norm Perspective. This viewpoint is formalized by the operator-norm framework of metrized deep learning (Large et al., 2024; Bernstein and Newhouse, 2025), where each layer is equipped with a geometry derived from its input-output semantics, and gradients, treated as dual vectors, are transported back to the primal weight space by a layerwise duality map (Nemirovski and Yudin, 1983; Beck and Teboulle, 2003). For MLP layers under a rescaled spectral norm, this map recovers Muon’s orthogonalization (Bernstein and Newhouse, 2024; Jordan et al., 2024b). Both sides of the spectral condition then live in this layerwise operator norm: the size prescriptions on $\mathbf{W}$ and $\Delta\mathbf{W}$, as well as the steepest-descent rule that produces compatible updates. Steepest descent, however, only bounds $\|\Delta\mathbf{W}\|_{S_\infty}$ per step, leaving the control of $\|\mathbf{W}\|_{S_\infty}$ along the trajectory as a separate problem.

Spectral Control. Existing mechanisms for keeping $\|\mathbf{W}\|_{S_\infty}$ bounded during training fall into three broad families. *Decoupled weight decay* shrinks $\mathbf{W}$ uniformly at every step. Chen et al. (2025) cast Muon with decoupled weight decay as a member of the Lion- $\mathcal{K}$ family (Chen et al., 2024) whose trajectories converge to KKT points of a spectral-norm-constrained problem, with related Frank–Wolfe interpretations developed in (Sfyraki and Wang, 2025; Pethick et al., 2025). *Spectral reparameterizations* hardwire the singular spectrum of $\mathbf{W}$, e.g. as orthogonal under Parseval/Stiefel parameterizations (Cisse et al., 2017). *Spectral-norm caps* act on the top singular value via soft penalties (Yoshida and Miyato, 2017), post-hoc normalization (Miyato et al., 2018), or odd-polynomial clipping (Newhouse et al., 2025). All three families act on the singular spectrum of $\mathbf{W}$ as an aggregate quantity, leaving open which component of $\mathbf{W}$ empirically drives spectral-norm growth under Muon.

Our work answers this open question at the parameterization level. We identify the maximal row magnitude $\|\mathbf{g}_t\|_\infty$ as the empirical driver of $\|\mathbf{W}_t\|_{S_\infty}$ drift under Muon (Section 3.1) and reparameterize $\mathbf{W}$ so that this driver is promoted to an explicit optimizer variable, updated under its natural ℓ_∞ geometry. Spectral-norm control thus emerges from the construction of the

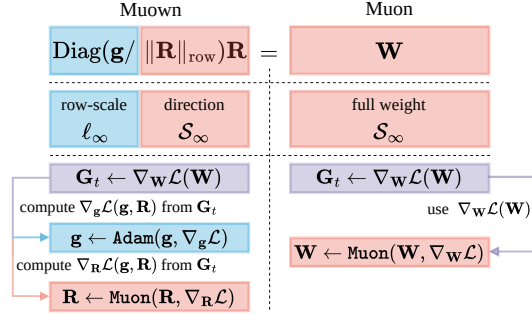


Figure 2: Abstract Illustration of Muown and Muon, showing the parameterization (top), semantics and geometry (middle), and the optimizer step (bottom).

parameterization rather than from an aggregate constraint on $\mathbf{W}$, avoiding both the indiscriminate shrinkage of decoupled weight decay and the orthogonality prior of spectral reparameterizations.

3 METHOD

Notation. Let $\mathbf{W}_t \in \mathbb{R}^{m \times n}$ denote the current iterate of a weight matrix of a linear layer with m and n output and input neurons, respectively, and $\mathbf{G}_t = \nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W}_t) \in \mathbb{R}^{m \times n}$ its corresponding gradient at iteration t with respect to a scalar loss $\mathcal{L}(\cdot)$. Division by vectors is elementwise. We write $\text{diag}(\mathbf{A}) \in \mathbb{R}^k$ for the vector of diagonal entries of a square matrix $\mathbf{A} \in \mathbb{R}^{k \times k}$, and $\text{Diag}(\mathbf{a}) \in \mathbb{R}^{k \times k}$ for the diagonal matrix with diagonal given by $\mathbf{a} \in \mathbb{R}^k$. For $p \in [1, \infty]$, $\|\mathbf{a}\|_p$ denotes the ℓ_p -norm of a vector $\mathbf{a} \in \mathbb{R}^k$, and $\|\mathbf{A}\|_{S_p}$ the Schatten- p norm of a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, i.e. the ℓ_p -norm of its singular values $\sigma_1(\mathbf{A}) \geq \dots \geq \sigma_{\min(m,n)}(\mathbf{A}) \geq 0$. In particular $\|\mathbf{A}\|_{S_\infty}$ is the spectral norm and $\|\mathbf{A}\|_{S_1}$ the nuclear norm. The canonical (Frobenius) inner product on $\mathbb{R}^{m \times n}$ is $\langle \mathbf{A}, \mathbf{B} \rangle := \text{tr}(\mathbf{A}^\top \mathbf{B})$ with induced norm $\|\mathbf{A}\|_F$, and $\langle a, b \rangle = a^\top b$ on $\mathbb{R}^k$. We write $\lambda_{\max}(\mathbf{M})$ for the largest eigenvalue of a symmetric matrix $\mathbf{M}$. With slight abuse of notation, we denote the row-norm vector $\|\mathbf{A}\|_{\text{row}} = (\|\mathbf{A}_{1,:}\|_2, \dots, \|\mathbf{A}_{m,:}\|_2)^\top \in \mathbb{R}^m$. For $\mathbf{A} \in \mathbb{R}^{m \times n}$, we define the projection which removes the per-row radial component with respect to $\mathbf{X} \in \mathbb{R}^{m \times n}$ as $\text{Proj}_{\mathbf{X}}(\mathbf{A}) := \mathbf{A} - \text{Diag}(\text{diag}(\mathbf{A}\mathbf{X}^\top)) \mathbf{X}$. In other words, for each row $\mathbf{a}_i$, $\text{Proj}_{\mathbf{X}}(\mathbf{A})$ subtracts $\langle \mathbf{a}_i, \mathbf{x}_i \rangle \mathbf{x}_i$.

3.1 MOTIVATION

Row Norm Drift. The constrained optimization and metrized-deep-learning arguments in Section 2 single out $\|\mathbf{W}_t\|_{S_\infty}$ as the quantity that governs stability of feature propagation, yet steepest spectral descent only bounds $\|\Delta \mathbf{W}_t\|_{S_\infty}$ per step. Before prescribing a separate control mechanism for $\|\mathbf{W}_t\|_{S_\infty}$ itself, we ask a more basic question: *which component of $\mathbf{W}_t$ is responsible for the observed spectral-norm drift under Muon?* The following decomposition isolates two candidate mechanisms.

Proposition 1 (Spectral Norm Decomposition into Row Magnitude and Coherence). *Let $\mathbf{W}_t \in \mathbb{R}^{m \times n}$ and suppose each row of $\mathbf{W}_t$ is nonzero. Define the row-magnitude vector $\mathbf{g}_t := \|\mathbf{W}_t\|_{\text{row}} \in \mathbb{R}^m$, and the row-normalized matrix $\mathbf{D}_t := \text{Diag}(1/\mathbf{g}_t) \mathbf{W}_t \in \mathbb{R}^{m \times n}$, so that $\mathbf{W}_t = \text{Diag}(\mathbf{g}_t) \mathbf{D}_t$. Let $\mathbf{C}_t := \mathbf{D}_t \mathbf{D}_t^\top \in \mathbb{R}^{m \times m}$ be the Gram matrix of the unit-norm rows, and define $\mathbf{p}_t := |\mathbf{g}_t| / \|\mathbf{g}_t\|_\infty \in [0, 1]^m$ with $\mathbf{P}_t := \text{Diag}(\mathbf{p}_t)$. Then the spectral norm of $\mathbf{W}_t$ admits the decomposition*

$$\|\mathbf{W}_t\|_{S_\infty}^2 = \|\mathbf{g}_t\|_\infty^2 \lambda_{\max}(\mathbf{P}_t \mathbf{C}_t \mathbf{P}_t).$$

In particular, the spectral norm splits into a row-scale term $\|\mathbf{g}_t\|_\infty^2$ and a normalized alignment term $\lambda_{\max}(\mathbf{P}_t \mathbf{C}_t \mathbf{P}_t)$.

The proof is deferred to Appendix B.1. We remark that the two factors are *disjoint* by design: any positive uniform row-rescaling of $\mathbf{W}_t$ moves only $\|\mathbf{g}_t\|_\infty$, while any directional change to the rows moves only $\lambda_{\max}(\mathbf{P}_t \mathbf{C}_t \mathbf{P}_t)$. The alignment factor $\lambda_{\max}(\mathbf{P}_t \mathbf{C}_t \mathbf{P}_t) \geq 1$ measures coherence of the unit-norm neurons in $\mathbf{D}_t$, weighted by the normalized row-scale profile $\mathbf{p}_t$. It collapses to 1 if and only if the (non-negligible) rows are orthonormal, in which case $\|\mathbf{W}_t\|_{S_\infty} = \|\mathbf{g}_t\|_\infty$ and the layer’s operator norm is determined entirely by its maximal row scale.

Row Norm Control. Across the attention and MLP linear layers of a 500M model trained with Muon, we find that $\|\mathbf{W}_t\|_{S_\infty}$ and $\|\mathbf{g}_t\|_\infty$ move in lock-step throughout training (Fig. 1a, 1b; Fig. 10, 11), whereas $\lambda_{\max}(\mathbf{P}_t \mathbf{C}_t \mathbf{P}_t)$ only exhibits a brief transient rise early in training, remaining bounded thereafter (Fig. 1c, 1d; Fig. 12). A simple causal intervention confirms that the row scales drive the spectral norm drift: freezing $\mathbf{g}_t = \mathbf{g}_1$ after initialization, a minimal intervention that touches only the row-scale factor, eliminates the systematic growth of $\|\mathbf{W}_t\|_{S_\infty}$ (Figs. 1, 10).

With $\|\mathbf{g}_t\|_\infty$ identified as the empirical driver of $\|\mathbf{W}_t\|_{S_\infty}$ under Muon, we respond by promoting it from a byproduct of the $\mathbf{W}$ -update to a trainable variable, optimized under its own geometry. Existing spectral-control schemes instead intervene at coarser granularities of Proposition 1. Trainable row magnitudes, by contrast, act only on the identified driver $\|\mathbf{g}_t\|_\infty$, without the

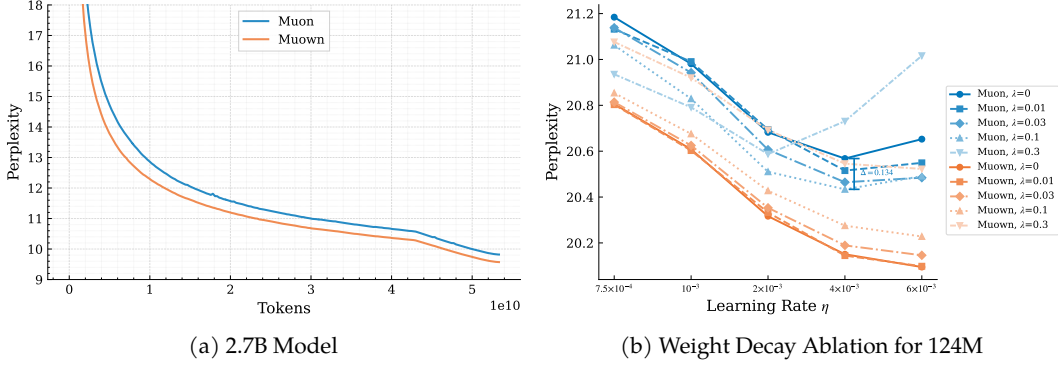


Figure 4: *Left*: Perplexity of Muon and Muown on a 2.7B transformer trained on FineWeb-Edu. *Right*: Ablation of weight decay and learning rate for Muon and Muown on 124M models after 5B tokens.

orthogonality prior of reparameterizations and without the indiscriminate shrinkage of weight decay.

3.2 MUOWN

To turn $\|\mathbf{g}_t\|_\infty$ into a controllable variable, we reparameterize each linear layer as $\mathbf{W}(\mathbf{g}, \mathbf{R}) = \text{Diag}(\mathbf{g} / \|\mathbf{R}\|_{\text{row}}) \mathbf{R}$ with $\mathbf{g} \in \mathbb{R}^m$ and $\mathbf{R} \in \mathbb{R}^{m \times n}$, i.e. weight normalization (Salimans and Kingma, 2016), under which $\mathbf{g}$ coincides exactly with the row-magnitude vector of Proposition 1. To remain agnostic to the model implementation, we realize the reparameterization *implicitly inside the optimizer*: $\mathbf{g}$ is held as an additional optimizer state, $\mathbf{R}$ is reconstructed from the stored effective weight $\mathbf{W}$ on each step, and the forward pass is left unchanged. The resulting procedure (Algorithm 1, Fig. 2) is therefore a drop-in replacement for Muon, which decomposes each layer into a direction $\mathbf{R}$, already metrized by the spectral norm, and a diagonal per-neuron gain $\text{Diag}(\mathbf{g})$.

It remains to choose a geometry for $\mathbf{g}$. Composability in the modular-duality framework (Bernstein and Newhouse, 2025, Def. 6) requires that the output space of $\mathbf{R}$ and the input space of $\text{Diag}(\mathbf{g})$ carry the same norm, so the natural metric on $\mathbf{g}$ is the induced operator norm $\|\text{Diag}(\Delta\mathbf{g})\|_{S_\infty}$. Since the operator norm of a diagonal matrix equals the largest entry in absolute value, this yields $\|\text{Diag}(\Delta\mathbf{g})\|_{S_\infty} = \|\Delta\mathbf{g}\|_\infty$, i.e., the ℓ_∞ geometry on $\mathbf{g}$. This is independently confirmed by Proposition 1, which identifies $\|\mathbf{g}_t\|_\infty$ as the row-scale factor of $\|\mathbf{W}_t\|_{S_\infty}$.

Steepest descent under ℓ_∞ is SignSGD (Bernstein et al., 2018), with the entrywise sign as the duality map (Bernstein and Newhouse, 2025, Ex. 2). Its update $\Delta\mathbf{g} = -\eta \text{sgn}(\nabla_{\mathbf{g}})$ satisfies $\|\Delta\mathbf{g}\|_\infty = \eta$ by construction, so the row-scale contribution $\|\text{Diag}(\Delta\mathbf{g})\|_{S_\infty} = \|\Delta\mathbf{g}\|_\infty$ to $\|\Delta\mathbf{W}_t\|_{S_\infty}$ is bounded at every step. In practice, SignSGD is the stateless analogue of Adam with both EMAs disabled (Balles et al., 2020; Xie and Li, 2024), and AdamW has been shown to converge to KKT points of an ℓ_∞ -constrained problem (Xie and Li, 2024). We therefore adopt Adam as the magnitude optimizer in Algorithm 1, as it inherits the ℓ_∞ geometry while adding momentum-based smoothing against stochastic-gradient noise.

We initialize $\mathbf{R}_1 \leftarrow \mathbf{W}_1$ and cached row norms and magnitudes as $\mathbf{r}, \mathbf{g}_1 \leftarrow \|\mathbf{W}_1\|_{\text{row}}$, so that $\mathbf{W}(\mathbf{g}_1, \mathbf{R}_1) = \mathbf{W}_1$, making Muown start from exactly the same point as Muon under any non-zero initialization scheme. All remaining optimizer states are zero. Following Liu et al. (2025), we scale the $\mathbf{R}$ -step by $0.2\sqrt{\max(m, n)}$ to match the RMS norm of a typical Adam update, so that a single learning rate η_t drives both the direction and magnitude updates and no separate learning rate for $\mathbf{g}$ is introduced. The resulting algorithm can be found in Algorithm 1.

4 CONVERGENCE ANALYSIS

We establish that Muown attains the optimal non-convex rate in the stochastic setting (Theorem 1) in a dual norm matching the two update geometries.

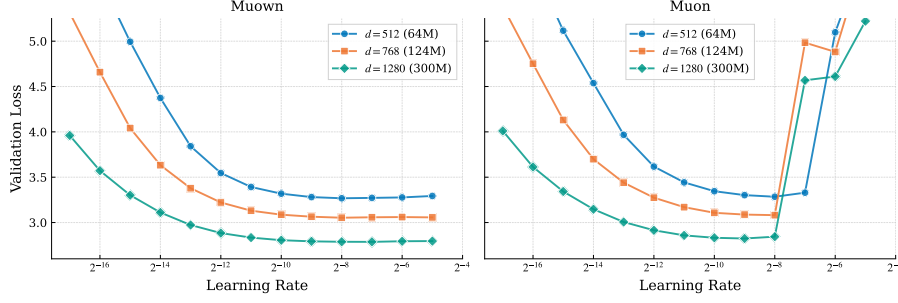


Figure 5: Validation loss of Muon and Muown at three different model widths of 512, 768, 1280 with 12 layers each. We observe that Muown yields close to optimal validation loss for a wide range of learning rates from 2^{-10} up to 2^{-5} , irrespective of model width.

Following the prevalent view of Adam as a smoothed sign method (Kunstner et al., 2023), we analyze SignSGD with momentum (Signum) as its proxy on g. SignSGD preserves the relevant ℓ_∞ geometry dynamics which largely explain Adam’s behavior in Transformer training (Kunstner et al., 2023), while making the underlying ℓ_∞ geometry explicit (Bernstein et al., 2018; Balles et al., 2020; Xie and Li, 2024).

The deterministic case follows and all proofs are deferred to Appendix B.

Setup and Geometry. Working in the re-parameterized variables $\mathbf{g} \in \mathbb{R}^m$, $\mathbf{R} \in \mathbb{R}_{\text{row}>0}^{m \times n}$, where $\mathbb{R}_{\text{row}>0}^{m \times n}$ denotes the set of $m \times n$ matrices with strictly positive row norms, we consider $\tilde{\mathcal{L}}: \mathbb{R}^m \times \mathbb{R}_{\text{row}>0}^{m \times n} \rightarrow \mathbb{R}$, $\tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}) := \mathcal{L}(\mathbf{W}(\mathbf{g}, \mathbf{R})) = \mathcal{L}\left(\text{Diag}\left(\frac{\mathbf{g}}{\|\mathbf{R}\|_{\text{row}}}\right) \mathbf{R}\right)$. Since the g- and R-updates take unit-sized steps in the ℓ_∞ - and spectral norms respectively, we equip the joint variable with the product max-norm $\|(\mathbf{g}, \mathbf{R})\| = \max\{\|\mathbf{g}\|_\infty, \|\mathbf{R}\|_{S_\infty}\}$ and the canonical inner product $\langle(\mathbf{g}, \mathbf{R}), (\mathbf{g}', \mathbf{R}')\rangle := \langle\mathbf{g}, \mathbf{g}'\rangle + \langle\mathbf{R}, \mathbf{R}'\rangle$. The corresponding dual norm is $\|(\mathbf{g}, \mathbf{R})\|_* = \|\mathbf{g}\|_1 + \|\mathbf{R}\|_{S_1}$, combining the ℓ_1 -norm on $\mathbb{R}^m$ with the Schatten-1 (nuclear) norm on $\mathbb{R}^{m \times n}$. Each summand is the dual of the norm in which the corresponding update is bounded, so the product norm simply lifts the modular-duality viewpoint of Section 2 to the joint variable $(\mathbf{g}, \mathbf{R})$.

Stochastic Convergence. We consider the objective $\mathcal{L}(\mathbf{W}) := \mathbb{E}_\xi[\mathcal{L}(\mathbf{W}; \xi)]$ with access only to stochastic gradients $\nabla \mathcal{L}(\mathbf{W}, \xi)$, and use the re-parametrized gradient oracles $\nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t, \xi_t) = \text{diag}(\nabla \mathcal{L}(\mathbf{W}(\mathbf{g}_t, \mathbf{R}_t), \xi_t) \mathbf{D}_t^\top)$ and $\nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t, \xi_t) = \text{Diag}\left(\frac{\mathbf{g}_t}{\|\mathbf{R}_t\|_{\text{row}}}\right) \text{Proj}_{\mathbf{D}_t}(\nabla \mathcal{L}(\mathbf{W}(\mathbf{g}_t, \mathbf{R}_t), \xi_t))$, where $\mathbf{D}_t = \text{Diag}(1/\|\mathbf{R}_t\|_{\text{row}}) \mathbf{R}_t$ is the row-normalization of $\mathbf{R}_t$.

Theorem 1 (Stochastic Convergence of Muown). *Assume $\tilde{\mathcal{L}}$ is L -smooth along the trajectory, lower bounded by $\tilde{\mathcal{L}}_*$ and the gradient oracles have $\sigma_{\mathbf{g}}$ and $\sigma_{\mathbf{R}}$ bounded variance accordingly (see Assumption 1). Denote $\Delta_1 \geq \tilde{\mathcal{L}}(\mathbf{g}_1, \mathbf{R}_1) - \tilde{\mathcal{L}}_*$ and the norm equivalence constants $\zeta_{\mathbf{g}} := \max_{\mathbf{g}} \|\mathbf{g}\|_1 / \|\mathbf{g}\|_2$, $\zeta_{\mathbf{R}} := \max_{\mathbf{R}} \|\mathbf{R}\|_{S_1} / \|\mathbf{R}\|_{\text{F}}$. Then the iterates of Algorithm 2 with $\eta_t \equiv \sqrt{\frac{\Delta_1(1-\beta_1)}{LT}}$ and $\beta_1 = 1 -$*

Algorithm 1 Step of Muon with integrated Weight Normalization (Muown)

Require: $\mathbf{W} \in \mathbb{R}^{m \times n}$, gradient $\nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W}) \in \mathbb{R}^{m \times n}$, states $\mathbf{g}, \mathbf{r}, \mathbf{m}_{\mathbf{g}}, \mathbf{v}_{\mathbf{g}} \in \mathbb{R}^m$, $\mathbf{M} \in \mathbb{R}^{m \times n}$, learning rate η_t , weight decay λ , momentum β_1

$\mathbf{R} \leftarrow \text{Diag}(\frac{\mathbf{r}}{\mathbf{g}}) \mathbf{W}$
 $\mathbf{D} \leftarrow \text{Diag}(1/\mathbf{r}) \mathbf{R}$
obtain decoupled gradients
 $\nabla_{\mathbf{g}} \mathcal{L}(\mathbf{W}) \leftarrow (\nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W}) \odot \mathbf{D}) \mathbf{1}_n$
 $\nabla_{\mathbf{R}} \mathcal{L}(\mathbf{W}) \leftarrow \text{Diag}(\frac{\mathbf{g}}{\mathbf{r}}) \text{Proj}_{\mathbf{D}}(\nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W}))$
run muon w/ nesterov on direction
 $\mathbf{M} \leftarrow \beta_1 \mathbf{M} + \nabla_{\mathbf{R}} \mathcal{L}(\mathbf{W})$
 $\mathbf{O} \leftarrow \arg \min_{\|\mathbf{O}\|_{S_\infty} \leq 1} \langle \beta_1 \mathbf{M} + \nabla_{\mathbf{R}} \mathcal{L}(\mathbf{W}), \mathbf{O} \rangle$
 $\mathbf{R} \leftarrow \mathbf{R} + 0.2 \sqrt{\max(m, n)} \eta_t \mathbf{O}$
run adam on magnitude
 $\mathbf{g} \leftarrow \text{Adam}(\nabla_{\mathbf{g}} \mathcal{L}(\mathbf{W}), \mathbf{m}_{\mathbf{g}}, \mathbf{v}_{\mathbf{g}}, \eta_t)$
obtain effective weight
 $\mathbf{r} \leftarrow \|\mathbf{R}\|_{\text{row}}$
 $\mathbf{W} \leftarrow \text{Diag}(\frac{\mathbf{g}}{\mathbf{r}}) \mathbf{R}$

$$\min \left\{ 1, \max \left\{ T^{-2/3}, \sqrt{\frac{\Delta_1 L}{\hat{\sigma}^2 T}} \right\} \right\}, \text{ where } \hat{\sigma} := \zeta_{\mathbf{g}} \sigma_{\mathbf{g}} + \zeta_{\mathbf{R}} \sigma_{\mathbf{R}}, \text{ satisfy}$$

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\left\| \nabla \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_* \right] \leq 6 \sqrt{\frac{\Delta_1 L}{T}} + 8 \left(\frac{\Delta_1 L \hat{\sigma}^2}{T} \right)^{1/4} + \frac{4\hat{\sigma}}{T^{1/3}}$$

which matches the optimal $\mathcal{O} \left(\left(\frac{\Delta_1 L \sigma^2}{T} \right)^{1/4} \right)$ non-convex rate up to constants.

The guarantee is stated for the dual norm of the re-parameterized gradient, and the dominant term matches the optimal stochastic non-convex rate up to constants (Arjevani et al., 2023), mirroring the rate established for Muon by (Shen et al., 2025, Corollary 4.4). Moreover, since $\|\cdot\|_1 \geq \|\cdot\|_2$ and $\|\cdot\|_{S_1} \geq \|\cdot\|_F$, the bound is strictly stronger than its ℓ_2 /Frobenius counterpart. We caution that the bound concerns $\nabla \tilde{\mathcal{L}}$. Although $\tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}) = \mathcal{L}(\mathbf{W})$ pointwise, $\nabla \tilde{\mathcal{L}}$ and $\nabla \mathcal{L}$ need not be directly comparable. The only algorithm-dependent factor is the noise coefficient $\hat{\sigma} = \zeta_{\mathbf{g}} \sigma_{\mathbf{g}} + \zeta_{\mathbf{R}} \sigma_{\mathbf{R}}$, where the norm-equivalence constants $\zeta_{\mathbf{g}}, \zeta_{\mathbf{R}}$ arise, as in recent Muon analyses (Shen et al., 2025, Theorem 4.3), from controlling the stochastic error in ℓ_1 /Schatten-1 from ℓ_2 /Frobenius variance bounds. In comparison, the corresponding factor for Muon is $\zeta_{\mathbf{W}} \sigma_{\mathbf{W}}$, hence our theory predicts an improvement of Muown over Muon whenever $\zeta_{\mathbf{g}} \sigma_{\mathbf{g}} + \zeta_{\mathbf{R}} \sigma_{\mathbf{R}} < \zeta_{\mathbf{W}} \sigma_{\mathbf{W}}$. Section 5 and Fig. 7 verify this inequality empirically throughout training, with the gap widening up to a factor of $1.6\times$, offering reduced gradient noise in the $(\mathbf{g}, \mathbf{R})$ -parameterization as a candidate mechanism for explaining the observed perplexity gains.

In the proof, we first use our choice of geometry to split the descent Lemma into independent $\mathbf{g}$ - and $\mathbf{R}$ -terms (Lemma 1). We then apply the steepest-descent analyses of SignSGD and Muon, and recombine without cross-terms. The full proof can be found in Appendix B.2.

5 EXPERIMENTS

In this section, we evaluate Muown on language-model pre-training with modern GPT-style architectures on FineWeb-Edu. This section is designed to answer (i) whether Muown can improve convergence relative to current state-of-the-art geometry-aware optimizers at a fixed token budget, (ii) whether it reduces sensitivity to learning rate and weight decay tuning, and (iii) whether its convergence behavior is better conditioned, as indicated by gradient rank and gradient noise. We conclude this section with a discussion of the computational overhead.

Setup. We adopt the experimental setup of Ajroldi (2024) which is based on a nanoGPT (Karpathy, 2022) implementation augmented with recent architectural improvements such as RoPE (Su et al., 2024), RMSNorm normalization (Zhang and Sennrich, 2019), and SwiGLU activations (Shazeer, 2020). We use a warmup-stable-decay learning rate schedule (Hu et al., 2024) and train most models at four sizes: 124M, 500M, 1B, and 2.7B. We focus on perplexity at a fixed token budget as a primary evaluation criterion. All training details can be found in Appendix C.

Table 1: Perplexity for Muown, Muon, and SOAP on a 500M model for different learning rates. We report average and standard deviation across 3 seeds.

η	Muown	Muon		SOAP
	$\lambda = 0.0$	$\lambda = 0.1$	$\lambda = 0.0$	$\lambda = 10^{-4}$
4×10^{-3}	14.13 $\pm$ 0.004	28.37 $\pm$ 23.59	88.15 $\pm$ 3.869	14.53 $\pm$ 0.017
2×10^{-3}	14.14 $\pm$ 0.027	14.37 $\pm$ 0.028	14.40 $\pm$ 0.042	14.47 $\pm$ 0.022
1×10^{-3}	14.24 $\pm$ 0.026	14.44 $\pm$ 0.039	14.54 $\pm$ 0.041	14.49 $\pm$ 0.036

Pre-training Perplexity. We organize pre-training results by scale: a full learning-rate sweep at 124M (Fig. 3), a multi-seed comparison at 500M (Table 1), and single-seed runs at 1B and 2.7B (Table 2, Fig. 4a). Unless explicitly stated otherwise, Muown is run without weight decay ($\lambda = 0$). For Muon we use $\lambda = 0.1$. Both of these we find to be optimal across $\lambda \in \{0, 0.01, 0.03, 0.1, 0.3\}$ on the 124M grid (Fig. 4b; discussed in detail below). We additionally include $\lambda = 0$ for Muon.

At the 124M scale, Fig. 3 compares Muown against Muon, SOAP, and Lion across eight log-spaced learning rates, both with and without weight decay on the Muon-family optimizers. Muown attains the lowest perplexity at *every* learning rate in the grid and degrades gracefully at the top of the range, whereas Muon diverges for learning rates at which Muown is still near-optimal.

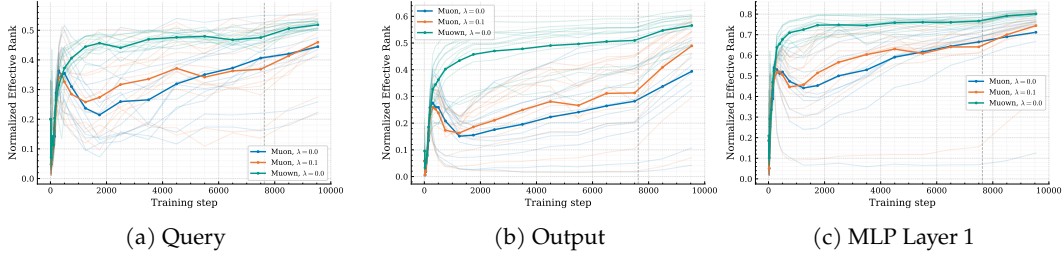


Figure 6: Normalized effective rank (Roy and Vetterli, 2007) of minibatch gradients for a 124M model. At a given training step, we sample 8 gradients and average their effective ranks.

At 500M (Table 1), Muown retains at least a 0.2 perplexity improvement over the best Muon configuration at every matched learning rate, significantly larger than the per-seed standard deviation of stable non-divergent runs. The stability gap widens at $\eta = 4 \times 10^{-3}$: Muon diverges (88.15 at $\lambda = 0$) or exhibits large seed variance (28.37 ± 23.59 at $\lambda = 0.1$), while Muown gets closer to optimality, corroborating the robustness signature seen at 124M.

The gap persists at the 1B and 2.7B scale. At 1B (Table 2), Muown attains 11.90 perplexity at $\eta = 3 \times 10^{-3}$, improving upon the best Muon run (12.20 at $\eta = 2 \times 10^{-3}$, $\lambda = 0$) by 0.30. At 2.7B (Fig. 4a), Muown reaches 9.57 at $\eta = 2 \times 10^{-3}$ while the best stable Muon run lands at 9.82 ($\eta = 10^{-3}$, $\lambda = 0$), preserving a 0.25 perplexity advantage for Muown. The consistent perplexity gap from 500M up to 2.7B suggests that the improvement does not attenuate with scale within the regime we reach.

Two additional experiments probe whether these gains transfer beyond the architecture and the four baselines above. On a Qwen2-0.5B (Yang et al., 2024a) architecture (Appendix A.2), Muown preserves a 0.25 perplexity advantage over the best-tuned Muon configuration. Moreover, compared to NorMuon (Li et al., 2025), a per-row Muon variant that adapts the orthogonalized momentum via a row-wise second-moment statistic (Appendix A.3), Muown matches or improves perplexity in the near-optimal range and exhibits a wider basin of stability at 124M.

Robustness to Hyperparameter Choices.

We assess robustness along two axes: width-dependence of the optimal learning rate (Fig. 5) and sensitivity to weight decay (Fig. 4b). Following the protocol of Pethick et al. (2025), we sweep a log-spaced learning-rate grid from 2^{-16} to 2^{-5} across three widths of 512, 768, and 1280 (corresponding to 67M, 124M, and 308M parameters) at Chinchilla-optimal token count. Muown yields near-optimal validation loss over a wide plateau of learning rates spanning 2^{-10} to 2^{-5} at every width, whereas Muon’s optimal learning rate shifts downward with increasing width

and its near-optimal plateau narrows. This indicates that Muown slightly alleviates the need for per-scale learning-rate retuning. To rule out an implementation-specific artefact, we reproduce the same qualitative phenomenon using the widely-used modded-nanogpt (Jordan et al., 2024a) setup (Fig. 8).

Fig. 4b sweeps a two-dimensional grid of learning rate and weight decay $\lambda \in \{0, 0.01, 0.03, 0.1, 0.3\}$ for Muon and Muown at the 124M scale. At optimally tuned learning rate, weight decay provides a sizeable improvement in perplexity for Muon and is minimized at $\lambda = 0.1$, justifying the Muon baseline configuration used in the pre-training perplexity comparisons above. In contrast, the same sweep of weight decay strength yields no significant gain for Muown.

Table 2: Perplexity for Muown and Muon on 1B and 2.7B models across learning rates. We also report perplexity of AdamW at 1B. [†]Run aborted after 12h on 16 GPUs as validation loss was diverging.

Scale	η	Muown	Muon		AdamW
		$\lambda = 0$	$\lambda = 0.1$	$\lambda = 0$	$\lambda = 0.1$
1B	3×10^{-3}	11.90	44.89	110.19	303.16
	2×10^{-3}	11.93	12.33	12.20	12.69
	1×10^{-3}	12.01	12.27	12.32	12.93
	5×10^{-4}	12.18	12.35	12.44	12.85
2.7B	2×10^{-3}	9.57	— [†]	— [†]	
	1×10^{-3}	9.63	10.01	9.82	
	7.5×10^{-4}	9.66	9.96	9.87	

Effective Rank and Stochastic Noise. We probe two properties of the stochastic minibatch gradients each optimizer orthogonalizes, $\nabla_{\mathbf{W}}\mathcal{L}$ for Muon and $\nabla_{\mathbf{R}}\mathcal{L}$ for Muown, to investigate the perplexity gains reported above. First, following Ahn et al. (2025), we track the normalized effective rank $\text{erank}(\sigma)/\min(m, n)$ with $\text{erank}(\sigma) = \exp(-\sum_i p_i \log p_i)$, $p_i = \sigma_i/\|\sigma\|_1$ (Roy and Vetterli, 2007) which can be read as the fraction of directions the optimizer effectively explores. Fig. 6 shows that Muown sustains a higher effective rank than Muon on every reported weight type throughout training. Second, Section 4 showed that Muown’s stochastic rate scales with $\hat{\sigma} = \zeta_{\mathbf{g}}\sigma_{\mathbf{g}} + \zeta_{\mathbf{R}}\sigma_{\mathbf{R}}$ while Muon’s scales with $\zeta_{\mathbf{W}}\sigma_{\mathbf{W}}$ (Shen et al., 2025), leaving the theoretical edge conditional on $\hat{\sigma} < \zeta_{\mathbf{W}}\sigma_{\mathbf{W}}$. Fig. 7 verifies this inequality per weight matrix along the training trajectory: Muown’s noise term stays below Muon’s and the gap widens over training. Together, higher spectral diversity and lower stochastic noise in the $(\mathbf{g}, \mathbf{R})$ -parameterization offer a mechanistic explanation for the observed perplexity gains.

Resource Overhead. Table 3 reports median end-to-end step time on a 500M model (forward and backward passes, gradient synchronization, and optimizer update). Relative to Muon, Muown adds only element-wise $\mathcal{O}(mn)$ work per layer (the weight-norm decomposition and recomposition together with the Adam update on $\mathbf{g} \in \mathbb{R}^m$) asymptotically dominated by the shared Newton-Schulz iteration at $\mathcal{O}(m^2n)$ for $m \leq n$. We benchmark two variants: a *replicated* baseline that runs the optimizer step on every rank, and a *distributed* variant (under DDP) that follows Jordan et al. (2024a), sharding the per-layer optimizer work round-robin across ranks and concluding with an all-gather. In the replicated setting the surplus materializes as a measurable ≈ 20 ms (1.5%) per step; once sharded in the same pass as Newton-Schulz, step times for Muon and Muown become indistinguishable, with the gap shrinking further as the rank count grows. This amortization relies on exposing the reparameterization *inside* the optimizer: at the module level, every rank would redundantly recompute the decomposition on its local replica of $\mathbf{W}$ at every forward pass. Communication is unchanged from Muon: only $\mathbf{W}$ is all-gathered, while $\mathbf{g}$ and the reconstructed $\mathbf{R}$ remain rank-local.

Muown introduces *no* additional $\mathcal{O}(mn)$ state on top of Muon, since $\mathbf{R}$ is reconstructed inside the optimizer from the stored effective weight $\mathbf{W}$. The only new states are four $\mathbb{R}^m$ vectors per linear layer ($\mathbf{g}$, the cached row norm $\|\mathbf{R}\|_{\text{row}}$, and the Adam first- and second-moment buffers for $\mathbf{g}$) amounting to ≈ 2 MB of additional peak memory in the runs of Table 3, negligible next to Muon’s momentum buffer.

6 CONCLUSION

We show that the spectral-norm drift observed under Muon is not an irreducible property of the optimizer. Instead, the decomposition of Proposition 1 isolates the maximal row magnitude $\|\mathbf{g}_t\|_{\infty}$ as its empirical driver, while the row-coherence factor remains stable along the trajectory. We propose Muown, which acts on this driver at the parameterization level, lifting $\mathbf{g}$ to an optimizer variable updated under the ℓ_{∞} geometry singled out by modular duality and leaving Muon unchanged on the direction. We provide evidence that this intervention can improve perplexity, widen the plateau of near-optimal learning rates across model widths, and alleviate the need to tune weight decay. These gains come at a memory cost linear in the number of rows and negligible step-time overhead once the optimizer is sharded across ranks.

Limitations and Future Work. The decomposition $\mathbf{W} = \text{Diag}(\mathbf{g}/\|\mathbf{R}\|_{\text{row}})\mathbf{R}$ is undefined when a row of $\mathbf{W}$ is exactly zero at initialization, making the direction unidentifiable. Architectures that

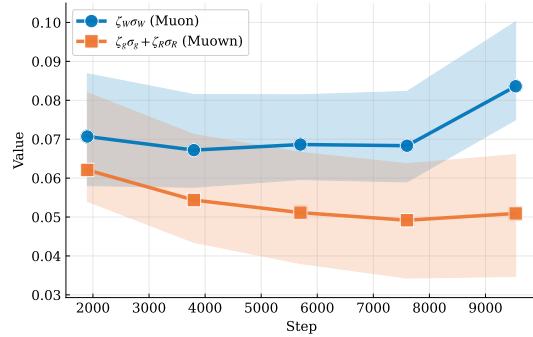


Figure 7: Comparison of the noise coefficients for Muon and Muown. Shaded areas mark the interquartile range over weights.

Table 3: Median step time (s) for Muown and Muon on a 500M model (1K steps, 4 GH200 GPUs).

Setting	Muon	Muown
Replicated	1.34	1.36
Distributed	1.27	1.27

depend on zero-initialized weight matrices, such as zero-initialized residual output projections or some LoRA adapters, therefore require either a small non-zero initialization or an opt-out from the reparameterization for the affected layers. Our empirical study is also confined to dense GPT-style architectures and pre-training horizons up to 53B tokens. The behavior on convolutional, mixture-of-experts, and substantially longer-horizon regimes remains open. A first natural direction for future work is to vary the geometry imposed on $\mathbf{g}$. Replacing ℓ_∞ with sparsity-inducing ℓ_1 -updates would transform magnitude control into structured row-wise sparsity, offering an optimizer-level route to neuron pruning and therefore potentially more efficient inference.

ACKNOWLEDGMENTS

This work was supported under project ID a0184 as part of the Swiss AI Initiative, through a small grant from the ETH Domain and computational resources provided by the Swiss National Supercomputing Centre (CSCS) under the Alps infrastructure. Kai Lion is supported by Swiss National Science Foundation (SNSF) Sinergia Funding No. 216600. Florian Hübler acknowledges financial support from the ETH research grant and Swiss National Science Foundation (SNSF) Project Funding No. 200021-207343, the Alexander von Humboldt Foundation, the European Union’s Horizon Europe research and innovation programme under grant agreements No. 101120237 (ELIAS), and No. 101070617 (ELSA). Views and opinions expressed are however those of the author only and do not necessarily reflect those of the European Union or European Commission. Neither the European Union nor the granting authority can be held responsible for them. Antonio Orvieto acknowledges the financial support of the Hector Foundation. Niao He is supported by an ETH research grant funded through the ETH Zurich Foundation and by an SNSF Starting Grant.

REFERENCES

- Kwangjun Ahn, Byron Xu, Natalie Abreu, and John Langford. Dion: Distributed Orthonormalized Updates, 2025. arXiv:2504.05295.
- Niccolò Ajroldi. plainLM: Language Model Pretraining in PyTorch. <https://github.com/Niccolo-Ajroldi/plainLM>, 2024.
- Yossi Arjevani, Yair Carmon, John C. Duchi, Dylan J. Foster, Nathan Srebro, and Blake Woodworth. Lower Bounds for Non-Convex Stochastic Optimization. *Mathematical Programming*, 199(1): 165–214, 2023.
- Lukas Balles, Fabian Pedregosa, and Nicolas Le Roux. The Geometry of Sign Gradient Descent, 2020. arXiv:2002.08056.
- Amir Beck and Marc Teboulle. Mirror Descent and Nonlinear Projected Subgradient Methods for Convex Optimization. *Operations Research Letters*, 31(3):167–175, 2003.
- Yoshua Bengio, Nicolas Boulanger-Lewandowski, and Razvan Pascanu. Advances in Optimizing Recurrent Networks. In *IEEE Trans. Acoust., Speech, Sig. Process.*, 2013.
- Jeremy Bernstein and Laker Newhouse. Old Optimizer, New Norm: An Anthology. In *OPT 2024: 16th Annual Workshop on Optimization for Machine Learning*, 2024.
- Jeremy Bernstein and Laker Newhouse. Modular Duality in Deep Learning. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2025.
- Jeremy Bernstein, Yu-Xiang Wang, Kamyar Azizzadenesheli, and Animashree Anandkumar. signSGD: Compressed Optimisation for Non-Convex Problems. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2018.
- Sidney Black, Stella Biderman, Eric Hallahan, Quentin Anthony, Leo Gao, Laurence Golding, Horace He, Connor Leahy, Kyle McDonell, Jason Phang, Michael Pieler, Usvsn Sai Prashanth, Shivanshu Purohit, Laria Reynolds, Jonathan Tow, Ben Wang, and Samuel Weinbach. GPT-NeoX-20B: An Open-Source Autoregressive Language Model. In *Proceedings of BigScience Episode #5 – Workshop on Challenges & Perspectives in Creating Large Language Models*. ACL, 2022.

-
- Léon Bottou, Frank E Curtis, and Jorge Nocedal. Optimization Methods for Large-scale Machine Learning. *SIAM review*, 60(2):223–311, 2018.
- Stephen Boyd and Lieven Vandenbergh. *Convex Optimization*. Cambridge University Press, 2004.
- David Carlson, Volkan Cevher, and Lawrence Carin. Stochastic Spectral Descent for Restricted Boltzmann Machines. In *Proc. Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, 2015.
- Lizhang Chen, Bo Liu, Kaizhao Liang, and Qiang Liu. Lion Secretly Solves Constrained Optimization: As Lyapunov Predicts. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2024.
- Lizhang Chen, Jonathan Li, and Qiang Liu. Muon Optimizes Under Spectral Norm Constraints. In *OPT 2025: 17th Annual Workshop on Optimization for Machine Learning*, 2025.
- Xiangning Chen, Chen Liang, Da Huang, Esteban Real, Kaiyuan Wang, Hieu Pham, Xuanyi Dong, Thang Luong, Cho-Jui Hsieh, Yifeng Lu, and Quoc V Le. Symbolic Discovery of Optimization Algorithms. In *Proc. Neural Information Processing Systems (NeurIPS)*, 2023.
- Moustapha Cisse, Piotr Bojanowski, Edouard Grave, Yann Dauphin, and Nicolas Usunier. Parseval Networks: Improving Robustness to Adversarial Examples. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2017.
- Ashok Cutkosky and Harsh Mehta. Momentum Improves Normalized SGD. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2020.
- Vineet Gupta, Tomer Koren, and Yoram Singer. Shampoo: Preconditioned Stochastic Tensor Optimization. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2018.
- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Oriol Vinyals, Jack W. Rae, and Laurent Sifre. Training Compute-optimal Large Language Models. In *Proc. Neural Information Processing Systems (NeurIPS)*, 2022.
- Shengding Hu, Yuge Tu, Xu Han, Chaoqun He, Ganqu Cui, Xiang Long, Zhi Zheng, Yewei Fang, Yuxiang Huang, Weilin Zhao, Xinrong Zhang, Zheng Leng Thai, Kaihuo Zhang, Chongyi Wang, Yuan Yao, Chenyang Zhao, Jie Zhou, Jie Cai, Zhongwu Zhai, Ning Ding, Chao Jia, Guoyang Zeng, Dahai Li, Zhiyuan Liu, and Maosong Sun. MiniCPM: Unveiling the Potential of Small Language Models with Scalable Training Strategies. In *Proc. Conf. on Language Modeling (COLM)*, 2024.
- Keller Jordan, Jeremy Bernstein, Brendan Rappazzo, @fernbear.bsky.social, Boza Vlado, You Jiacheng, Franz Cesista, Braden Koszarsky, and @Grad62304977. modded-nanoGPT: Speedrunning the NanoGPT Baseline. GitHub repository, 2024a. URL <https://github.com/KellerJordan/modded-nanogpt>.
- Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, and Jeremy Bernstein. Muon: An Optimizer for Hidden Layers in Neural Networks, 2024b. URL <https://kellerjordan.github.io/posts/muon/>.
- Andrej Karpathy. nanoGPT: The Simplest, Fastest Repository for Training/Finetuning Medium-sized GPTs. GitHub repository, 2022. URL <https://github.com/karpathy/nanoGPT>.
- Kimi-Team, Yifan Bai, Yiping Bao, Guanduo Chen, Jiahao Chen, Ningxin Chen, Ruijue Chen, and Yanru Chen et al. Kimi K2: Open Agentic Intelligence, 2025. arXiv:2507.20534.
- Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2015.
- Frederik Kunstner, Jacques Chen, Jonathan Wilder Lavington, and Mark Schmidt. Noise Is Not the Main Factor Behind the Gap Between Sgd and Adam on Transformers, But Sign Descent Might Be. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2023.

-
- Tim Large, Yang Liu, Minyoung Huh, Hyojin Bahng, Phillip Isola, and Jeremy Bernstein. Scalable Optimization in the Modular Norm. In *Proc. Neural Information Processing Systems (NeurIPS)*, 2024.
- Zichong Li, Liming Liu, Chen Liang, Weizhu Chen, and Tuo Zhao. NorMuon: Making Muon More Efficient and Scalable, 2025. arXiv:2510.05491.
- Jingyuan Liu, Jianlin Su, Xingcheng Yao, Zhejun Jiang, Guokun Lai, Yulun Du, Yidao Qin, Weixin Xu, Enzhe Lu, Junjie Yan, Yanru Chen, Huabin Zheng, Yibo Liu, Shaowei Liu, Bohong Yin, Weiran He, Han Zhu, Yuzhi Wang, Jianzhou Wang, Mengnan Dong, Zheng Zhang, Yongsheng Kang, Hao Zhang, Xinran Xu, Yutao Zhang, Yuxin Wu, Xinyu Zhou, and Zhilin Yang. Muon is Scalable for LLM Training, 2025. arXiv:2502.16982.
- Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2019.
- Anton Lozhkov, Loubna Ben Allal, Leandro von Werra, and Thomas Wolf. FineWeb-Edu: The Finest Collection of Educational Content, 2024. URL <https://huggingface.co/datasets/HuggingFaceFW/fineweb-edu>.
- James Martens and Roger Grosse. Optimizing Neural Networks with Kronecker-factored Approximate Curvature. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2015.
- Takeru Miyato, Toshiki Kataoka, Masanori Koyama, and Yuichi Yoshida. Spectral Normalization for Generative Adversarial Networks, 2018. arXiv:1802.05957.
- Arkadi Nemirovski and David B. Yudin. *Problem Complexity and Method Efficiency in Optimization*. Wiley-Interscience Series in Discrete Mathematics. Wiley-Interscience, 1983.
- Yu.E. Nesterov. A Method for Solving the Convex Programming Problem with Convergence Rate $\mathcal{O}(1/K^2)$. *Doklady Akademii Nauk SSSR*, 269(3):543–547, 1983. ISSN 0002-3264.
- Laker Newhouse, R. Preston Hess, Franz Cesista, Andrii Zahorodnii, Jeremy Bernstein, and Phillip Isola. Training Transformers with Enforced Lipschitz Constants, 2025. arXiv:2507.13338.
- Thomas Pethick, Wanyun Xie, Kimon Antonakopoulos, Zhenyu Zhu, Antonio Silveti-Falls, and Volkan Cevher. Training Deep Learning Models with Norm-Constrained LMOs. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2025.
- Olivier Roy and Martin Vetterli. The Effective Rank: A Measure of Effective Dimensionality. In *15th European Signal Processing Conference*, pages 606–610. IEEE, 2007.
- Tim Salimans and Durk P Kingma. Weight Normalization: A Simple Reparameterization to Accelerate Training of Deep Neural Networks. In *Proc. Neural Information Processing Systems (NeurIPS)*, 2016.
- Maria-Eleni Sfyraiki and Jun-Kun Wang. Lions and Muons: Optimization via Stochastic Frank-Wolfe, 2025. arXiv:2506.04192.
- Noam Shazeer. GLU Variants Improve Transformer, 2020. arXiv:2002.05202.
- Wei Shen, Ruichuan Huang, Minhui Huang, Cong Shen, and Jiawei Zhang. On the Convergence Analysis of Muon, 2025. arXiv:2505.23737.
- Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. RoFormer: Enhanced Transformer with Rotary Position Embedding. *Neurocomput.*, 2024.
- Mark Tuddenham, Adam Prügel-Bennett, and Jonathan Hare. Orthogonalising gradients to speed up neural network optimisation, 2022. arXiv:2202.07052.
- Nikhil Vyas, Depen Morwani, Rosie Zhao, Itai Shapira, David Brandfonbrener, Lucas Janson, and Sham Kakade. SOAP: Improving and Stabilizing Shampoo using Adam. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2025.

Shuo Xie and Zhiyuan Li. Implicit Bias of AdamW: ℓ_∞ Norm Constrained Optimization, 2024. arXiv:2404.04454.

An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, Guanting Dong, Haoran Wei, Huan Lin, Jialong Tang, Jialin Wang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Ma, Jianxin Yang, Jin Xu, Jingren Zhou, Jinze Bai, Jinzheng He, Junyang Lin, Kai Dang, Keming Lu, Keqin Chen, Kexin Yang, Mei Li, Mingfeng Xue, Na Ni, Pei Zhang, Peng Wang, Ru Peng, Rui Men, Ruize Gao, Runji Lin, Shijie Wang, Shuai Bai, Sinan Tan, Tianhang Zhu, Tianhao Li, Tianyu Liu, Wenbin Ge, Xiaodong Deng, Xiaohuan Zhou, Xingzhang Ren, Xinyu Zhang, Xipin Wei, Xuancheng Ren, Xuejing Liu, Yang Fan, Yang Yao, Yichang Zhang, Yu Wan, Yunfei Chu, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, Zhifang Guo, and Zhihao Fan. Qwen2 Technical Report, 2024a. arXiv:2407.10671.

Greg Yang, James B. Simon, and Jeremy Bernstein. A Spectral Condition for Feature Learning, 2024b. arXiv:2310.17813.

Yuichi Yoshida and Takeru Miyato. Spectral Norm Regularization for Improving the Generalizability of Deep Learning, 2017. arXiv:1705.10941.

Biao Zhang and Rico Sennrich. Root Mean Square Layer Normalization. In *Proc. Neural Information Processing Systems (NeurIPS)*, 2019.

Supplementary Document for Muown

CONTENTS

1	Introduction	1
1.1	Contribution	2
2	Related Work	2
3	Method	4
3.1	Motivation	4
3.2	Muown	5
4	Convergence Analysis	5
5	Experiments	7
6	Conclusion	9
A	Additional Experiments	15
A.1	nanoGPT Speedrunning	15
A.2	Qwen2-0.5B	15
A.3	Comparison with NorMuon	15
A.4	Spectral Analysis	16
A.5	Magnitude Learning	16
A.6	Effective Rank of Minibatch Gradients	17
B	Proofs	18
B.1	Proposition 1	18
B.2	Deterministic Convergence (Theorem 2)	19
B.3	Stochastic Convergence (Theorem 1)	21
C	Experimental Details	24

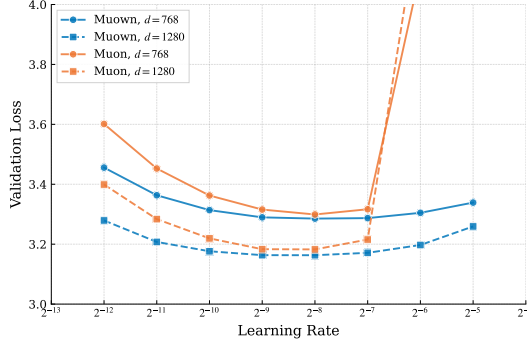


Figure 8: Learning rate and width ablation for Muon and Muown using the modded-nanoGPT codebase.

A ADDITIONAL EXPERIMENTS

A.1 NANO GPT SPEEDRUNNING

We compare Muown and Muon for different learning rates and widths using the modded-nanoGPT codebase and report the results in Fig. 8. Similar to Pethick et al. (2025), we choose a log-space learning grid from 2^{-12} to 2^{-5} . Fig. 8 confirms the conclusions drawn in the main text regarding the greater robustness with respect to ill-chosen learning rates.

A.2 QWEN2-0.5B

As an additional architectural datapoint, we pre-train a Qwen2-0.5B (Yang et al., 2024a) model, reporting results in Table 4. We remark that for all but one learning rate considered, Muown outperforms Muon and maintains a 0.25 perplexity advantage when comparing the best-performing hyperparameters.

Table 4: Perplexity for Muown and Muon on Qwen2-0.5B pre-training.

η	Muown	Muon	
	$\lambda = 0.0$	$\lambda = 0.1$	$\lambda = 0.0$
1×10^{-2}	14.57	—	—
8×10^{-3}	14.59	46.77	94.02
4×10^{-3}	14.65	14.86	14.83
2×10^{-3}	14.76	14.82	14.88
1×10^{-3}	14.95	14.91	15.03

A.3 COMPARISON WITH NORMUON

Both NorMuon (Li et al., 2025) and Muown augment plain Muon with a per-row mechanism on the rows of the 2D weight matrices, but they intervene at different points in the update. Plain Muon orthogonalizes the momentum of $\nabla_{\mathbf{W}} \mathcal{L}$ via Newton-Schulz and applies it directly to the parameter without a row-wise state. NorMuon instead applies a post-processing step to the update, maintaining a per-row EMA of the mean-squared entries of the orthogonalized update, dividing the update element-wise by its square root, and renormalizing back to the original Frobenius norm. This procedure effectively redistributes mass of the update across rows. In contrast, rather than redistributing mass of the update Muown changes the parameterization to control the row-norms of the weight. In essence, both methods equip every row with a scalar quantity (a second-moment statistic for NorMuon and a learned magnitude for Muown). Nevertheless, these row variables serve different purposes. We compare

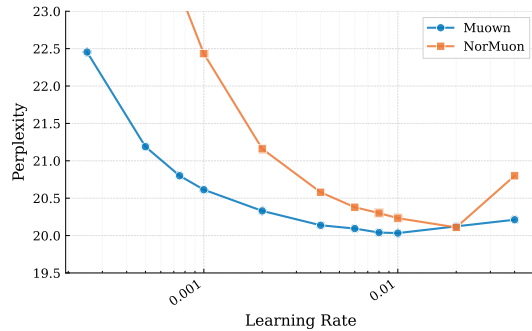


Figure 9: Comparison of NorMuon and Muown on a 124M model.

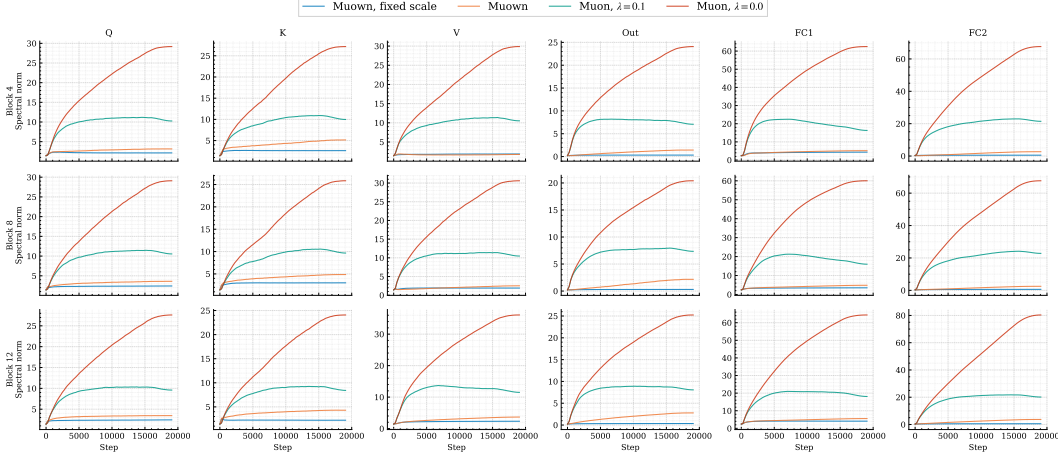


Figure 10: $\|\mathbf{W}_t\|_{S_\infty}$ for the layers in the 4th, 8th, and 12th transformer block of a 500M model.

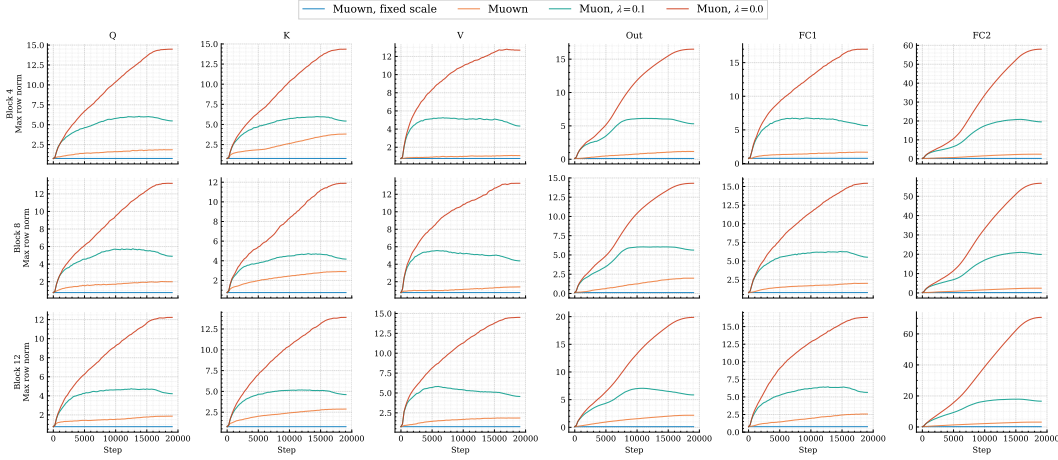


Figure 11: $\|\mathbf{g}_t\|_\infty$ for the layers in the 4th, 8th, and 12th transformer block of a 500M model.

both methods on a 124M model across eleven learning rates spanning 2.5×10^{-4} to 4×10^{-2} . Muown matches or improves on NorMuon at most learning rates that lie near the optimum and exhibits a noticeably wider basin of stability.

A.4 SPECTRAL ANALYSIS

We provide more detailed results for the empirical association between $\|\mathbf{W}_t\|_2^2$ and $\|\mathbf{g}_t\|_\infty$. Figs. 10, 11, and 12, show the evolution of $\|\mathbf{W}_t\|_2^2$, $\|\mathbf{g}_t\|_\infty$, and $\lambda_{\max}(\mathbf{P}_t \mathbf{C}_t \mathbf{P}_t)$, respectively. Each row depicts the quantities mentioned above computed on the weights of a transformer block of a 500M model trained with either of Muon and Muown. We remark that for all layers and weight types considered the spectral norm and maximal row norm are moving in lock step, while $\lambda_{\max}(\mathbf{P}_t \mathbf{C}_t \mathbf{P}_t)$ remains bounded.

Moreover, we provide more detailed results on the spectral distribution of the layers in the 16th transformer block in Figs. 13, 14, and 15.

A.5 MAGNITUDE LEARNING

In Section 3.1, we identify $\|\mathbf{g}_t\|_\infty$ as the empirical driver of $\|\mathbf{W}_t\|_{S_\infty}$ drift under Muon and remark that freezing $\mathbf{g}_t = \mathbf{g}_1$ at initialization eliminates the systematic growth in spectral norm. Nevertheless, it remains open how effective such a strategy is in terms of perplexity. As Table 5 demonstrates

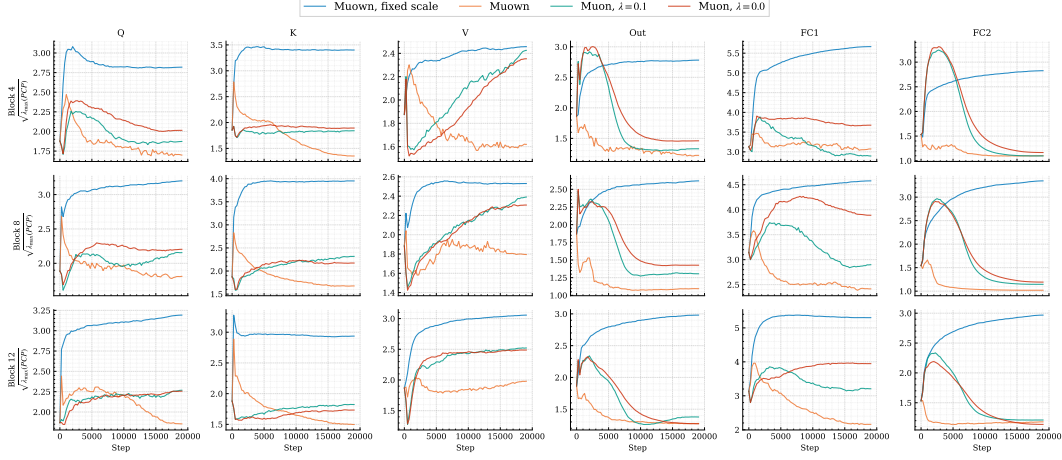


Figure 12: $\sqrt{\lambda_{\max}(\mathbf{P}_t \mathbf{C}_t \mathbf{P}_t)}$ for the layers in the 4th, 8th, and 12th transformer block of a 500M model.

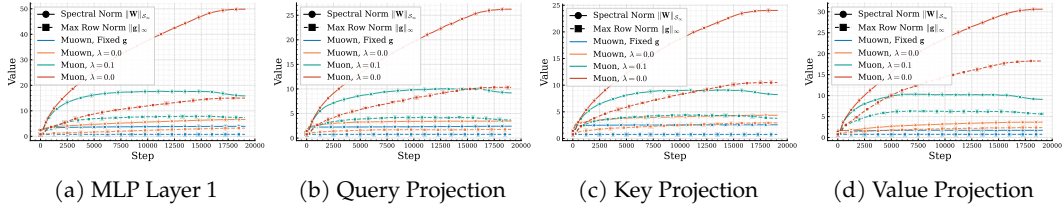


Figure 13: Evolution of spectral norm $\|\mathbf{W}_t\|_{S_\infty}$ and maximum row norm $\|\mathbf{g}_t\|_\infty$ over the course of training for linear layers of the 16th transformer block of a 500M model.

empirically, suppressing row-norm growth alone already accounts for a significant part of the improvement over Muon: *Fixed*, which freezes row magnitudes at initialization, outperforms Muon at the smaller learning rates and prevents the divergence at $\eta = 4 \times 10^{-3}$ (Table 5), without learning the magnitudes. Training *g* on top, with either of SignSGD+Momentum (Signum) or Adam, refines the result further, with the ordering between Adam, Signum, and Fixed preserved across all three learning rates. While we find the *Fixed* variant to perform slightly worse than the ones with trainable magnitudes, it can serve as a more lightweight alternative that does not require storage of the Adam states and further simplifies the update procedure.

A.6 EFFECTIVE RANK OF MINIBATCH GRADIENTS

Combined with Fig. 6 in the main text, Fig. 16 covers all six linear weight types of the transformer block (queries, keys, values, output projection, and both MLP layers). The same pattern holds across types: Muown sustains a higher normalized effective rank than Muon throughout training on every weight type considered, with the gap opening early in training and persisting for the entire run.

Table 5: Perplexity for Muown on a 500M model using different magnitude optimizer choices.

η	Muown		
	Fixed	Signum	Adam
4×10^{-3}	14.22	14.21	14.11
2×10^{-3}	14.21	14.18	14.11
1×10^{-3}	14.30	14.27	14.24

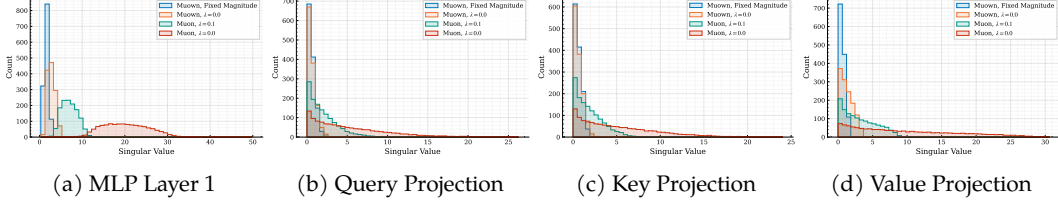


Figure 14: Histogram of singular values for linear layers of the 16th transformer block of a 500M model.

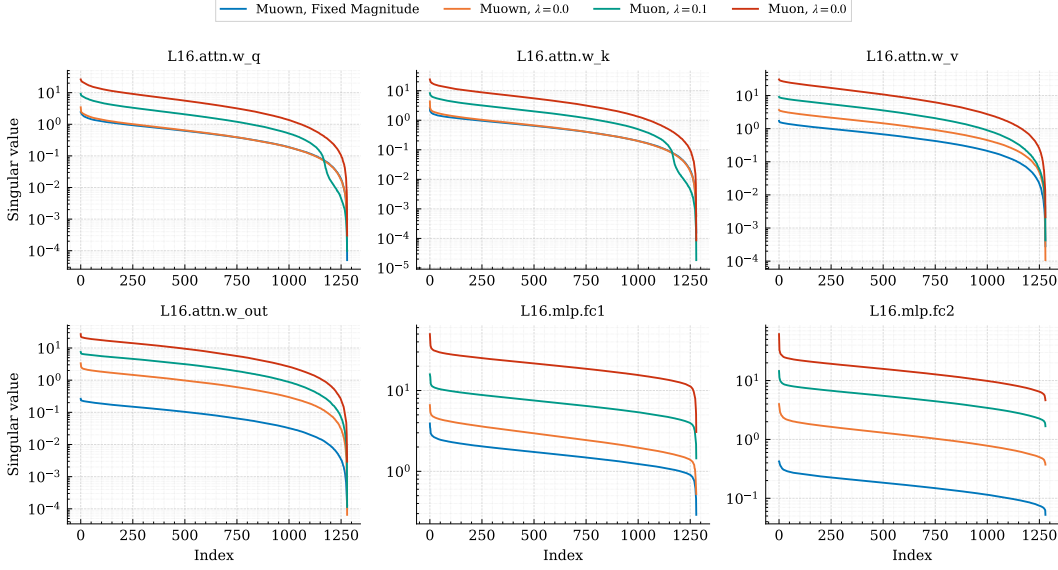


Figure 15: Singular value distribution for linear layers of the 16th transformer block of a 500M model. attn refers to the attention block with w_q , w_k , w_v , w_{out} denoting query, key, value, and output projections, respectively. fc1 and fc2 refer to the first and second MLP layers of the block.

B PROOFS

B.1 PROPOSITION 1

Proof. Since each row of $\mathbf{W}_t$ is nonzero, the vector $\mathbf{g}_t = \|\mathbf{W}_t\|_{\text{row}}$ is well-defined and has strictly positive entries. Hence

$$\mathbf{D}_t = \text{Diag}(1/\mathbf{g}_t) \mathbf{W}_t$$

is well-defined, and for each row i ,

$$\mathbf{d}_{t,i} = \frac{\mathbf{w}_{t,i}}{\|\mathbf{w}_{t,i}\|_2},$$

so $\|\mathbf{d}_{t,i}\|_2 = 1$. Therefore,

$$\mathbf{W}_t = \text{Diag}(\mathbf{g}_t) \mathbf{D}_t.$$

Using this factorization, we compute

$$\mathbf{W}_t \mathbf{W}_t^\top = \text{Diag}(\mathbf{g}_t) \mathbf{D}_t \mathbf{D}_t^\top \text{Diag}(\mathbf{g}_t) = \text{Diag}(\mathbf{g}_t) \mathbf{C}_t \text{Diag}(\mathbf{g}_t).$$

By definition of the spectral norm,

$$\|\mathbf{W}_t\|_{S_\infty}^2 = \lambda_{\max}(\mathbf{W}_t \mathbf{W}_t^\top),$$

and therefore

$$\|\mathbf{W}_t\|_{S_\infty}^2 = \lambda_{\max}(\text{Diag}(\mathbf{g}_t) \mathbf{C}_t \text{Diag}(\mathbf{g}_t)).$$

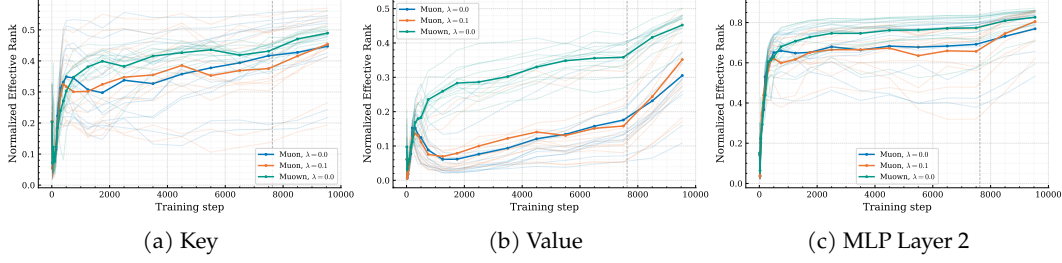


Figure 16: Normalized effective rank (Roy and Vetterli, 2007) of minibatch gradients for a 124M model. At a given training step, we sample 8 gradients and average their effective ranks.

Algorithm 2 Muown (SignSGD Version)

Require: Initial point $\mathbf{W}_1 \in \mathbb{R}^{m \times n}$ with nonzero rows; stepsizes $\eta, \gamma > 0$; momentum $\beta_1 \in [0, 1]$
Initialize $\mathbf{R}_1 \leftarrow \mathbf{W}_1$, $\mathbf{g}_1 \leftarrow \|\mathbf{W}_1\|_{\text{row}}$, $\mathbf{M}_0 \leftarrow \nabla_{\mathbf{R}} \mathcal{L}(\mathbf{W}_1, \xi_1)$, and $\mathbf{m}_0 \leftarrow \nabla_{\mathbf{g}} \mathcal{L}(\mathbf{W}_1, \xi_1)$.
for $t = 1, 2, \dots$ **do**
 # Update $\mathbf{R}_t$ with Muon
 $\mathbf{M}_t \leftarrow \beta_1 \mathbf{M}_{t-1} + \nabla_{\mathbf{R}} \mathcal{L}(\mathbf{W}_t, \xi_t)$
 $\mathbf{O}_t \leftarrow \arg \min_{\|\mathbf{O}\|_{S_\infty} \leq 1} \langle \mathbf{M}_t, \mathbf{O} \rangle$ ▷ Choose arbitrary for set-valued arg min.
 $\mathbf{R}_{t+1} \leftarrow \mathbf{R}_t + \eta \mathbf{O}_t$

 # Update $\mathbf{g}_t$ with SignSGD
 $\mathbf{m}_t \leftarrow \beta_1 \mathbf{m}_{t-1} + \nabla_{\mathbf{g}} \mathcal{L}(\mathbf{W}_t, \xi_t)$
 $\mathbf{g}_{t+1} \leftarrow \mathbf{g}_t - \gamma \text{sgn}(\mathbf{m}_t)$

 # Update $\mathbf{W}_t$
 $\mathbf{W}_{t+1} \leftarrow \mathbf{W}(\mathbf{g}_{t+1}, \mathbf{R}_{t+1})$
end for

It remains to factor out the maximal row magnitude. Since $\mathbf{g}_t = \|\mathbf{W}_t\|_{\text{row}}$, its entries are nonnegative, so $|\mathbf{g}_t| = \mathbf{g}_t$. By the definition of $\mathbf{p}_t$,

$$\mathbf{g}_t = \|\mathbf{g}_t\|_\infty \mathbf{p}_t, \quad \text{hence} \quad \text{Diag}(\mathbf{g}_t) = \|\mathbf{g}_t\|_\infty \mathbf{P}_t.$$

Substituting this into the previous identity yields

$$\text{Diag}(\mathbf{g}_t) \mathbf{C}_t \text{Diag}(\mathbf{g}_t) = \|\mathbf{g}_t\|_\infty^2 \mathbf{P}_t \mathbf{C}_t \mathbf{P}_t.$$

Since scaling a matrix by a positive scalar scales all eigenvalues by the same scalar, we obtain

$$\lambda_{\max}(\text{Diag}(\mathbf{g}_t) \mathbf{C}_t \text{Diag}(\mathbf{g}_t)) = \|\mathbf{g}_t\|_\infty^2 \lambda_{\max}(\mathbf{P}_t \mathbf{C}_t \mathbf{P}_t).$$

Recalling that the LHS of the equation above is equal to $\|\mathbf{W}_t\|_{S_\infty}^2$ proves the claim. $\square$

B.2 DETERMINISTIC CONVERGENCE (THEOREM 2)

In this section we will provide the convergence analysis of Algorithm 2. Remember that

$$\tilde{\mathcal{L}}: \mathbb{R}^m \times \mathbb{R}_{\text{row} > 0}^{m \times n} \rightarrow \mathbb{R}, \quad \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}) := \mathcal{L}(\mathbf{W}(\mathbf{g}, \mathbf{R})) = \mathcal{L}\left(\text{Diag}\left(\frac{\mathbf{g}}{\|\mathbf{R}\|_{\text{row}}}\right) \mathbf{R}\right).$$

Theorem 2 (Deterministic Convergence of Muown). *Assume $\tilde{\mathcal{L}}$ is L -smooth along the trajectory and lower bounded by $\tilde{\mathcal{L}}_\star$. Denote $\Delta_1 \geq \tilde{\mathcal{L}}(\mathbf{g}_1, \mathbf{R}_1) - \tilde{\mathcal{L}}_\star$ and assume the deterministic setting, i.e., access to*

the true gradient $\nabla \mathcal{L}(\mathbf{W})$. Then, with $\beta_1 = 0$ and $\eta = \gamma = \sqrt{\frac{\Delta_1}{LT}}$, the iterates of Algorithm 2 satisfy

$$\frac{1}{T} \sum_{t=1}^T \left\| \nabla \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_* \leq 4\sqrt{\frac{L\Delta_1}{T}}.$$

This matches the optimal $\mathcal{O}\left(\sqrt{L\Delta_1/T}\right)$ non-convex rate (up to constants).

We will derive this result in steps. Straightforward calculations show that the gradient of $\tilde{\mathcal{L}}$ with respect to the canonical inner product is given by

$$\begin{aligned} \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}) &= \text{diag}(\nabla \mathcal{L}(\mathbf{W}) \mathbf{D}^\top), \\ \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}) &= \text{Diag}(\mathbf{g} / \|\mathbf{R}\|_{\text{row}}) \left(\nabla \mathcal{L}(\mathbf{W}) - \text{Diag}(\nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R})) \mathbf{D} \right). \end{aligned}$$

where $\mathbf{D} = \text{Diag}(1/\|\mathbf{R}\|_{\text{row}})\mathbf{R}$ is the row-normalization of $\mathbf{R}$ and $\mathbf{W} = \text{Diag}(\mathbf{g})\mathbf{D}$ is the original parameterization. In particular we note that, after re-parameterization, Muown updates $\mathbf{R}$ and $\mathbf{g}$ with standard, albeit different, first-order updates. In particular Algorithm 2 corresponds exactly to Muown when replacing Adam with its proxy SignSGD. In a first step we show that, under a smoothness assumption on $\tilde{\mathcal{L}}$, the mixed update of Algorithm 2 can be analyzed separately and combined without issues.

Lemma 1. Let $\mathbf{g}, \Delta \mathbf{g} \in \mathbb{R}^m$, $\mathbf{R} \in \mathbb{R}_{\text{row}>0}^{m \times n}$, $\Delta \mathbf{R} \in \mathbb{R}^{m \times n}$ and $\gamma, \eta > 0$. Furthermore define $\mathbf{g}^+ := \mathbf{g} + \gamma \Delta \mathbf{g}$, $\mathbf{R}^+ := \mathbf{R} + \eta \Delta \mathbf{R}$ and assume $\mathbf{R}^+ \in \mathbb{R}_{\text{row}>0}^{m \times n}$ and that $\tilde{\mathcal{L}}$ is L -smooth between $(\mathbf{g}, \mathbf{R})$ and $(\mathbf{g}^+, \mathbf{R}^+)$. Then

$$\begin{aligned} &\tilde{\mathcal{L}}(\mathbf{g}^+, \mathbf{R}^+) - \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}) \\ &\leq \underbrace{\gamma \langle \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}), \Delta \mathbf{g} \rangle + \frac{L\gamma^2}{2} \|\Delta \mathbf{g}\|_\infty^2}_{\text{only depends on } \Delta \mathbf{g}} + \underbrace{\eta \langle \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}), \Delta \mathbf{R} \rangle_{\text{F}} + \frac{L}{2} \eta^2 \|\Delta \mathbf{R}\|_{S_\infty}^2}_{\text{only depends on } \Delta \mathbf{R}}. \end{aligned}$$

Proof. By the L -smoothness of $\tilde{\mathcal{L}}$ we have

$$\begin{aligned} &\tilde{\mathcal{L}}(\mathbf{g}^+, \mathbf{R}^+) - \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}) \\ &\leq \langle \nabla \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}), (\mathbf{g}^+, \mathbf{R}^+) - (\mathbf{g}, \mathbf{R}) \rangle_{\text{F}} + \frac{L}{2} \|(\mathbf{g}^+, \mathbf{R}^+) - (\mathbf{g}, \mathbf{R})\|^2 \\ &= \gamma \langle \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}), \Delta \mathbf{g} \rangle + \eta \langle \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}), \Delta \mathbf{R} \rangle_{\text{F}} + \frac{L}{2} \max\{\gamma^2 \|\Delta \mathbf{g}\|_\infty^2, \eta^2 \|\Delta \mathbf{R}\|_{S_\infty}^2\} \\ &\leq \gamma \langle \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}), \Delta \mathbf{g} \rangle + \frac{L\gamma^2}{2} \|\Delta \mathbf{g}\|_\infty^2 + \eta \langle \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}, \mathbf{R}), \Delta \mathbf{R} \rangle_{\text{F}} + \frac{L}{2} \eta^2 \|\Delta \mathbf{R}\|_{S_\infty}^2, \end{aligned}$$

where we used our choice of inner product and norm in the equality. $\square$

Now we are ready to prove Theorem 2.

Proof of Theorem 2. Let $t \in [T]$. We apply Lemma 1 with

$$\Delta \mathbf{g}_t := -\text{sgn}(\nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t)) \quad \text{and} \quad \Delta \mathbf{R}_t := \text{argmin}_{\|\mathbf{O}\|_{S_\infty} \leq 1} \langle \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), \mathbf{O} \rangle.$$

to get

$$\begin{aligned} &\tilde{\mathcal{L}}(\mathbf{g}_{t+1}, \mathbf{R}_{t+1}) - \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \\ &\leq \underbrace{\gamma \langle \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), \Delta \mathbf{g}_t \rangle + \frac{L\gamma^2}{2} \|\Delta \mathbf{g}_t\|_\infty^2}_{\text{only depends on } \Delta \mathbf{g}_t} + \underbrace{\eta \langle \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), \Delta \mathbf{R}_t \rangle_{\text{F}} + \frac{L}{2} \eta^2 \|\Delta \mathbf{R}_t\|_{S_\infty}^2}_{\text{only depends on } \Delta \mathbf{R}_t} \end{aligned}$$

For the $\Delta \mathbf{g}_t$ -term note that $\langle \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), \Delta \mathbf{g}_t \rangle = -\left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_1$ and $\|\Delta \mathbf{g}_t\|_\infty \leq 1$, thus

$$\gamma \langle \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), \Delta \mathbf{g}_t \rangle + \frac{L\gamma^2}{2} \|\Delta \mathbf{g}_t\|_\infty^2 \leq -\gamma \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_1 + \frac{L\gamma^2}{2}.$$

For the $\Delta \mathbf{R}_t$ -term we have $\left\langle \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), \Delta \mathbf{R}_t \right\rangle_{\text{F}} = -\left\| \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_{S_1}$, where $\|\cdot\|_{S_1}$ is the Schatten-1 (trace) norm and $\|\Delta \mathbf{R}_t\|_{S_\infty} \leq 1$, thus

$$\eta \left\langle \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), \Delta \mathbf{R}_t \right\rangle_{\text{F}} + \frac{L}{2} \eta^2 \|\Delta \mathbf{R}_t\|_{S_\infty}^2 \leq -\eta \left\| \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_{S_1} + \frac{L}{2} \eta^2.$$

Combining these bounds gives

$$\tilde{\mathcal{L}}(\mathbf{g}_{t+1}, \mathbf{R}_{t+1}) - \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \leq -\gamma \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_1 - \eta \left\| \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_{S_1} + \frac{L}{2} (\eta^2 + \gamma^2).$$

Summing this inequality from $t = 1$ to T and using $\tilde{\mathcal{L}}(\mathbf{g}_{T+1}, \mathbf{R}_{T+1}) \geq \tilde{\mathcal{L}}_\star$ gives

$$\sum_{t=1}^T \left(\gamma \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_1 + \eta \left\| \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_{S_1} \right) \leq \Delta_1 + \frac{L}{2} (\eta^2 + \gamma^2) T.$$

Define $R_T := \Delta_1 + \frac{L}{2} (\eta^2 + \gamma^2) T$. Since both terms in the sum are nonnegative, we can drop one term at a time to obtain

$$\frac{1}{T} \sum_{t=1}^T \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_1 \leq \frac{R_T}{\gamma T}, \quad \frac{1}{T} \sum_{t=1}^T \left\| \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_{S_1} \leq \frac{R_T}{\eta T}.$$

By our choice of product norm $\|(\mathbf{g}, \mathbf{R})\| = \max\{\|\mathbf{g}\|_\infty, \|\mathbf{R}\|_{S_\infty}\}$, the corresponding dual norm satisfies

$$\left\| \nabla \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_* = \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_1 + \left\| \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_{S_1}.$$

Combining the two bounds yields

$$\frac{1}{T} \sum_{t=1}^T \left\| \nabla \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_* \leq \frac{R_T}{\gamma T} + \frac{R_T}{\eta T}.$$

□

B.3 STOCHASTIC CONVERGENCE (THEOREM 1)

For the stochastic convergence result, we will use the following standard bounded variance assumption on the stochastic gradients along the trajectory of Algorithm 2.

Assumption 1. *The stochastic gradients along the trajectory are unbiased, i.e.,*

$$\mathbb{E}[\nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t, \xi_t)] = \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), \quad \mathbb{E}[\nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t, \xi_t)] = \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t),$$

and have bounded variance

$$\mathbb{E} \left[\left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t, \xi_t) - \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_1^2 \right] \leq \sigma_{\mathbf{g}}^2, \quad \mathbb{E} \left[\left\| \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t, \xi_t) - \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_{S_1}^2 \right] \leq \sigma_{\mathbf{R}}^2.$$

Now we are ready to prove the Theorem.

Proof. As in the deterministic case, we apply Lemma 1 to get

$$\begin{aligned} & \tilde{\mathcal{L}}(\mathbf{g}_{t+1}, \mathbf{R}_{t+1}) - \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \\ & \leq \gamma \langle \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), -\text{sgn}(\mathbf{m}_t) \rangle + \eta \left\langle \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), \mathbf{O}_t \right\rangle_{\text{F}} + \frac{L}{2} (\gamma^2 + \eta^2). \end{aligned} \tag{1}$$

We start by bounding the first term on the right-hand side through the classical decomposition of the inner product (see e.g. [Cutkosky and Mehta, 2020](#)) to get

$$\begin{aligned}
& \left\langle \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), -\text{sgn}(\mathbf{m}_t) \right\rangle \\
&= \left\langle \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) - (1 - \beta_1) \mathbf{m}_t, -\text{sgn}(\mathbf{m}_t) \right\rangle + \langle (1 - \beta_1) \mathbf{m}_t, -\text{sgn}(\mathbf{m}_t) \rangle \\
&\leq \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) - (1 - \beta_1) \mathbf{m}_t \right\|_1 \left\| -\text{sgn}(\mathbf{m}_t) \right\|_\infty - (1 - \beta_1) \left\| \mathbf{m}_t \right\|_1 \\
&\leq \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) - (1 - \beta_1) \mathbf{m}_t \right\|_1 - (1 - \beta_1) \left\| \mathbf{m}_t \right\|_1 \\
&\leq 2 \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) - (1 - \beta_1) \mathbf{m}_t \right\|_1 - \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_1,
\end{aligned}$$

where we applied the Hölder inequality in the second, and the triangle inequality in the fourth step. Further, when denoting $\boldsymbol{\mu}_t := \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) - (1 - \beta_1) \mathbf{m}_t$, $\boldsymbol{\epsilon}_t := \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t, \xi_t) - \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t)$,

$$\boldsymbol{\mu}_t = \beta_1^{t-1} \boldsymbol{\mu}_1 - (1 - \beta_1) \sum_{\tau=1}^{t-1} \beta_1^{\tau-1} \boldsymbol{\epsilon}_{t-\tau+1} + \sum_{\tau=1}^{t-1} \beta_1^\tau \left(\nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_{t-\tau}, \mathbf{R}_{t-\tau}) - \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_{t+1-\tau}, \mathbf{R}_{t+1-\tau}) \right)$$

and hence

$$\begin{aligned}
\left\| \boldsymbol{\mu}_t \right\|_1 &\leq \beta_1^{t-1} \left\| \boldsymbol{\mu}_1 \right\|_1 + (1 - \beta_1) \left\| \sum_{\tau=1}^{t-1} \beta_1^{\tau-1} \boldsymbol{\epsilon}_{t-\tau+1} \right\|_1 \\
&\quad + \sum_{\tau=1}^{t-1} \beta_1^\tau \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_{t-\tau}, \mathbf{R}_{t-\tau}) - \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_{t+1-\tau}, \mathbf{R}_{t+1-\tau}) \right\|_1 \\
&\leq \beta_1^{t-1} \left\| \boldsymbol{\mu}_1 \right\|_1 + (1 - \beta_1) \left\| \sum_{\tau=1}^{t-1} \beta_1^{\tau-1} \boldsymbol{\epsilon}_{t-\tau+1} \right\|_1 + \gamma L \sum_{\tau=1}^{t-1} \beta_1^\tau \\
&\leq \beta_1^{t-1} \left\| \boldsymbol{\mu}_1 \right\|_1 + (1 - \beta_1) \left\| \sum_{\tau=1}^{t-1} \beta_1^{\tau-1} \boldsymbol{\epsilon}_{t-\tau+1} \right\|_1 + \frac{\gamma L}{1 - \beta_1}
\end{aligned} \tag{2}$$

Following existing analyses of Muon ([Shen et al., 2025](#)), we next use equivalence of norms in finite dimensional spaces to derive

$$\begin{aligned}
\mathbb{E} \left[\left\| \sum_{\tau=1}^{t-1} \beta_1^{\tau-1} \boldsymbol{\epsilon}_{t-\tau+1} \right\|_1 \right] &\leq \zeta_{\mathbf{g}} \mathbb{E} \left[\left\| \sum_{\tau=1}^{t-1} \beta_1^{\tau-1} \boldsymbol{\epsilon}_{t-\tau+1} \right\|_2 \right] \leq \zeta_{\mathbf{g}} \sqrt{\mathbb{E} \left[\left\| \sum_{\tau=1}^{t-1} \beta_1^{\tau-1} \boldsymbol{\epsilon}_{t-\tau+1} \right\|_2^2 \right]} \\
&= \zeta_{\mathbf{g}} \sqrt{\sum_{\tau=1}^{t-1} \beta_1^{2(\tau-1)} \mathbb{E} \left[\left\| \boldsymbol{\epsilon}_{t-\tau+1} \right\|_2^2 \right]} \leq \frac{\zeta_{\mathbf{g}} \sigma_{\mathbf{g}}}{\sqrt{1 - \beta_1^2}},
\end{aligned}$$

where we used the L^2 -orthogonality of the error terms in the second, and $\|\cdot\|_2 \leq \|\cdot\|_1$ in the last step. Plugging into (2) and summing yields

$$\begin{aligned}
\sum_{t=1}^T \mathbb{E} [\left\| \boldsymbol{\mu}_t \right\|_1] &\leq \sum_{t=1}^T \beta_1^{t-1} \mathbb{E} [\left\| \boldsymbol{\mu}_1 \right\|_1] + (1 - \beta_1) \sum_{t=1}^T \mathbb{E} \left[\left\| \sum_{\tau=1}^{t-1} \beta_1^{\tau-1} \boldsymbol{\epsilon}_{t-\tau+1} \right\|_1 \right] + \frac{\gamma L}{1 - \beta_1} T \\
&\leq \frac{1}{1 - \beta_1} \mathbb{E} [\left\| \boldsymbol{\mu}_1 \right\|_1] + \frac{\zeta_{\mathbf{g}} (1 - \beta_1) \sigma}{\sqrt{1 - \beta_1^2}} T + \frac{\gamma L}{1 - \beta_1} T \\
&\leq \frac{\sigma_{\mathbf{g}}}{1 - \beta_1} + \zeta_{\mathbf{g}} \sigma_{\mathbf{g}} \sqrt{1 - \beta_1} T + \frac{\gamma L}{1 - \beta_1} T,
\end{aligned}$$

where we used our choice of $\mathbf{m}_1$ in the last step. Applying the same arguments to the $\mathbf{R}$ -term in (1) yields

$$\sum_{t=1}^T \left\langle \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), \mathbf{O}_t \right\rangle \leq \frac{2\sigma_{\mathbf{R}}}{1 - \beta_1} + 2\zeta_{\mathbf{R}} \sigma_{\mathbf{R}} \sqrt{1 - \beta_1} T + \frac{2\eta L}{1 - \beta_1} T - \sum_{t=1}^T \left\| \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_{S_1}.$$

Summing (1) from $t = 1$ to T and using $\tilde{\mathcal{L}}(\mathbf{g}_{T+1}, \mathbf{R}_{T+1}) \geq \tilde{\mathcal{L}}_*$ gives

$$\begin{aligned} & \sum_{t=1}^T \mathbb{E} \left[\gamma \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_1 + \eta \left\| \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_{S_1} \right] \\ & \leq \Delta_1 + L \left(\frac{5\gamma^2}{2(1-\beta_1)} + \frac{5\eta^2}{2(1-\beta_1)} \right) T + 2(\zeta_{\mathbf{g}}\sigma_{\mathbf{g}} + \zeta_{\mathbf{R}}\sigma_{\mathbf{R}})\eta\sqrt{1-\beta_1}T + \frac{2(\sigma_{\mathbf{g}} + \sigma_{\mathbf{R}})\eta}{1-\beta_1}. \end{aligned}$$

By our choices of $\gamma = \eta$ and definition $\hat{\sigma} = \zeta_{\mathbf{g}}\sigma_{\mathbf{g}} + \zeta_{\mathbf{R}}\sigma_{\mathbf{R}}$, we can simplify

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\left\| \nabla \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_* \right] \leq \frac{\Delta_1}{\eta T} + \frac{5\eta L}{(1-\beta_1)} + 2\sqrt{1-\beta_1}\hat{\sigma} + 2\frac{\sigma_{\mathbf{g}} + \sigma_{\mathbf{R}}}{(1-\beta_1)T}, \quad (3)$$

where we used the fact that $\left\| \nabla \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_* = \left\| \nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_1 + \left\| \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_{S_1}$. Plugging our choice $\gamma = \eta = \sqrt{\frac{\Delta_1(1-\beta_1)}{LT}}$ in the above inequality yields

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\left\| \nabla \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_* \right] \leq 6\sqrt{\frac{\Delta_1 L}{(1-\beta_1)T}} + 2\sqrt{1-\beta_1}\hat{\sigma} + 2\frac{\sigma_{\mathbf{g}} + \sigma_{\mathbf{R}}}{(1-\beta_1)T}.$$

Further using the choice $\beta_1 = 1 - \min \left\{ 1, \max \left\{ T^{-2/3}, \sqrt{\frac{\Delta_1 L}{\hat{\sigma}^2 T}} \right\} \right\}$ gives

$$\begin{aligned} \sqrt{\frac{\Delta_1 L}{(1-\beta_1)T}} & \leq \sqrt{\frac{\Delta_1 L}{T}} + \left(\frac{\Delta_1 L \hat{\sigma}^2}{T} \right)^{1/4} \\ \sqrt{1-\beta_1}\hat{\sigma} & \leq \frac{\hat{\sigma}}{T^{1/3}} + \left(\frac{\Delta_1 L \hat{\sigma}^2}{T} \right)^{1/4} \\ \frac{\sigma_{\mathbf{g}} + \sigma_{\mathbf{R}}}{(1-\beta_1)T} & \leq \frac{\sigma_{\mathbf{g}} + \sigma_{\mathbf{R}}}{T^{1/3}} \leq \frac{\hat{\sigma}}{T^{1/3}} \end{aligned}$$

Combining these bounds yields

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\left\| \nabla \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t) \right\|_* \right] \leq 6\sqrt{\frac{\Delta_1 L}{T}} + 8 \left(\frac{\Delta_1 L \hat{\sigma}^2}{T} \right)^{1/4} + \frac{4\hat{\sigma}}{T^{1/3}}$$

and hence the result. $\square$

C EXPERIMENTAL DETAILS

Further Remarks on Algorithm 1. We use learning rate scaling to match the Adam update RMS norm (Liu et al., 2025) for Muon as well as Muown. This corresponds to a learning rate adjustment by a factor of $0.2\sqrt{\max(m, n)}$ for a layer of shape $m \times n$. Moreover, Muown as outlined in Algorithm 1 uses simplified Nesterov momentum (Nesterov, 1983; Bengio et al., 2013), matching the PyTorch implementation of Muon.

To keep Algorithm 1 concise, we describe the use of decoupled weight decay with Muown separately here. When decoupled weight decay with coefficient λ is enabled, we replace the last line of Algorithm 1 with $\mathbf{W} \leftarrow \text{Diag}(\frac{\mathbf{g}}{\mathbf{r}})\mathbf{R} - \eta_t\lambda\mathbf{W}$ and refresh $\mathbf{g} \leftarrow \|\mathbf{W}\|_{\text{row}}$ to preserve the invariant $\mathbf{g} = \|\mathbf{W}\|_{\text{row}}$. This is only one of several natural placements of weight decay under the $(\mathbf{g}, \mathbf{R})$ -parameterization. For instance, one could equally well decay only the direction component $\mathbf{R}$. The variant adopted above stays closest to the canonical decoupled formulation (Loshchilov and Hutter, 2019) on the effective weight $\mathbf{W}$ and is the one used throughout our experiments. A systematic study of other alternatives is left to future work.

General Setup. If not noted otherwise, we run all experiments on top of the implementation of Ajroldi (2024). We make our code available on github.¹ We train a transformer with Rotational Positional Embeddings (Su et al., 2024), RMSNorm (Zhang and Sennrich, 2019), SwiGLU MLPs (Shazeer, 2020), expansion ratio of 8/3, and weight tying. We employ a warmup-stable-decay schedule for 2% of training as warmup and 20% as cooldown. In all but the distributed benchmarking experiment in Table 3, we make use of the PyTorch implementation of Muon. We justify this deviation below. We employ batch size of 524k tokens per step training, training for a total token budget of $20 \times$ tokens/parameter (Hoffmann et al., 2022), if not otherwise noted. Regarding sequence length and training budget we make the following size-dependent choices:

- For models of the 124M class, we choose 12 layers with hidden dimension of 768 and 12 attention heads. We train with sequence length 1024 for 5B tokens and validate on 4M tokens. We remark that due to weight tying the actual number of parameters is 124M.
- For models with 500M parameters, we choose 22 layers with hidden dimension of 1280 and 20 attention heads. We train with sequence length 2048 for 10B tokens and validate on 10M tokens.
- For models with 1B parameters, we choose 24 layers with hidden dimension of 1792 and 28 heads. We train for 20B tokens with sequence length 2048, validating on 10M tokens.
- For models with 2.7B parameters, we choose 32 layers with dimension 2560 and 40 heads and train for 53B tokens, keeping everything else as in the 1B setup.

All models are trained on the 100B token sample of FineWeb-Edu (Lozhkov et al., 2024) and tokenized with the GPT-NeoX-20B tokenizer (Black et al., 2022), if not noted otherwise. Experiments are performed on either of NVIDIA GH200 and NVIDIA H100 GPUs and we estimate the total GPU hours spent as roughly $\approx 18\text{k}$ hours, including experimentation, development, and final runs.

Spectral Analysis (Fig. 1, 10, 11, 12, 14, 15). For the spectral analysis plots, we adopt the 500M setup outlined above and tune Muon and Muown with learning rate $\eta \in \{1 \times 10^{-3}, 2 \times 10^{-3}, 4 \times 10^{-3}\}$ and weight decay $\lambda \in \{0, 0.1\}$. We report results for the best runs in terms of validation loss, i.e. $\eta = 2 \times 10^{-3}$ with and without weight decay for Muon and $\eta = 4 \times 10^{-3}$ without weight decay for Muown. For the setup with fixed row magnitude, we re-use the best config from Muown.

Learning Rate Ablation (Fig. 3). For the ablation study in Fig. 3, we use a grid of $\eta \in \{2.5 \times 10^{-4}, 5 \times 10^{-4}, 7.5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}, 4 \times 10^{-3}, 6 \times 10^{-3}, 8 \times 10^{-3}\}$ for all optimizers apart from Lion. For Muon’s weight decay parameter, we use $\lambda = 0.1$, which we found to be optimal across a grid of $\lambda \in \{0.01, 0.03, 0.1, 0.3\}$ (see Fig. 4b). Similarly, based on the results of Fig. 4b, we choose $\lambda = 0.01$ for Muown when using weight decay, despite having only a marginal effect. For Lion, we use $\eta \in \{8 \times 10^{-5}, 10^{-4}, 2.5 \times 10^{-4}, 5 \times 10^{-4}, 7.5 \times 10^{-4}\}$, as Lion typically has a larger update norm than other optimizers, requiring a lower learning rate (Chen et al., 2023). This matches our observation of divergence for learning rates larger than these. Moreover,

¹<https://github.com/kcc-lion/muown>

Chen et al. (2023) prescribe a larger value for weight decay in the range of $3\text{-}10\times$ of the one used for AdamW. We opt for $5\times$ the optimal value of Muon, yielding $\lambda = 0.5$ for Lion. We retain the default values of $(\beta_1, \beta_2) = (0.9, 0.99)$ for Lion. For SOAP (Vyas et al., 2025), we use the default hyperparameters from the original implementation: $(\beta_1, \beta_2) = (0.9, 0.95)$, $\epsilon = 10^{-8}$, a preconditioner update frequency of 10 steps, and bias correction enabled. We set the maximum precondition dimension to 10,000, which excludes the embedding and head layers (vocabulary size 50,280) from preconditioning while covering all other layers. Weight decay for SOAP is set at $\lambda = 10^{-4}$, following Vyas et al. (2025).

2.7B Runs (Fig. 4a, Table 2). We adopt the 2.7B setup above and sweep $\eta \in \{7.5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$, with $\lambda = 0$ for Muown and $\lambda \in \{0, 0.1\}$ for Muon. The $\eta = 2 \times 10^{-3}$ Muon runs were aborted after 12h on 16 GPUs as the validation loss was diverging.

Weight Decay Ablation (Fig. 4b). We adopt the 124M setup described above and train for 5B tokens ($2\times$ Chinchilla-optimal), sweeping a two-dimensional grid over the learning rate $\eta \in \{7.5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}, 4 \times 10^{-3}, 6 \times 10^{-3}\}$ and weight decay $\lambda \in \{0, 0.01, 0.03, 0.1, 0.3\}$ for both Muon and Muown. We report perplexity at $2\times$ the Chinchilla-optimal token count (i.e. 5B tokens).

Learning Rate and Width Ablation (Fig. 5). We use the 124M architecture outlined above and vary the width as 512, 768, 1280, resulting in models of size 67M, 124M, and 308M. We train each size for a Chinchilla-optimal amount of tokens and use a logarithmically-spaced learning rate grid from 2^{-16} to 2^{-5} .

Gradient Conditioning (Fig. 6). The effective rank of a matrix is defined as $\text{erank}(\sigma) = \exp(-\sum_{i=1}^{\min(m,n)} p_i \log p_i)$ where $p_i = \frac{\sigma_i}{\|\sigma\|_1}$ and $\sigma \in \mathbb{R}^{\min(m,n)}$ contains the singular values of the gradient matrix. We normalize the effective rank by dividing it with $\min(m, n)$. For Muon, we show the gradient with respect to $\mathbf{W}$, and for Muown the gradient with respect to $\mathbf{R}$. Each line shows one of the 12 layers and the bold line the average.

Gradient Noise (Fig. 7). We base the gradient noise analysis on our 124M weight decay ablation setup and consider the configuration with the best final validation loss. This results in Muon ($\eta = 4 \times 10^{-3}, \lambda = 0.1$) and Muown without weight decay ($\eta = 6 \times 10^{-3}, \lambda = 0$) run. For each of these runs we save the model and optimizer states after $t = 1900, 3800, 5700, 7600$ and the final $T = 9538$ iterations. For every checkpoint, we first compute the true gradient $\nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W}_t)$ for Muon, and $\nabla_{\mathbf{g}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t), \nabla_{\mathbf{R}} \tilde{\mathcal{L}}(\mathbf{g}_t, \mathbf{R}_t)$ for Muown checkpoints over the whole 5B token training dataset $\{\xi_1, \dots, \xi_T\}$. In a second pass we then calculate, for each weight matrix $\mathbf{W}_t^{(i)}$, resp. $\mathbf{g}_t^{(i)}, \mathbf{R}_t^{(i)}$,

$$\sigma_{\mathbf{W}_t^{(i)}}^2 = \frac{1}{T} \sum_{\tau=1}^T \|\nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W}_t) - \nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W}_t, \xi_\tau)\|_{S_1}^2,$$

and similarly $\sigma_{\mathbf{R}_t^{(i)}}$ and $\sigma_{\mathbf{g}_t^{(i)}}$. The values of $\zeta_{\mathbf{W}}, \zeta_{\mathbf{g}}$ and $\zeta_{\mathbf{R}}$ are known analytically to be $\zeta_{\mathbf{W}} = \zeta_{\mathbf{R}} = \sqrt{\min\{m, n\}}$ for $\mathbf{W}, \mathbf{R} \in \mathbb{R}^{m \times n}$ and $\zeta_{\mathbf{g}} = \sqrt{m}$ for $\mathbf{g} \in \mathbb{R}^m$. We compute the noise term separately for each weight matrix, plot the median across weights as a bold line, and show the interquartile range as a shaded region. We find that the noise term of Muown remains consistently below that of Muon, with the gap widening over the course of training.

nanoGPT Speedrunning (Fig. 8). For the nanoGPT speedrunning (Jordan et al., 2024a) experiments, we leverage the Muon record available on github.²

Multi-Seed Run for 500M (Table 1). We adopt the 500M setup above and report mean and standard deviation across three seeds for each configuration. The grid is $\eta \in \{1, 2, 4\} \times 10^{-3}$ for all optimizers. Muown is run with $\lambda = 0$; for Muon we report both $\lambda = 0$ and $\lambda = 0.1$, the latter identified as optimal in the 124M weight-decay ablation (Fig. 4b). SOAP follows its hyperparameters from the 124M learning-rate ablation above with $\lambda = 10^{-4}$.

Resource Overhead (Table 3). We compare the overall step times for Muon and Muown by running 1K steps for the 500M model outlined above on 4 GH200 devices. Here, we report the

²https://github.com/KellerJordan/modded-nanogpt/blob/2c7027462732f2f7a76fb4b021066442c7b29298/records/track_1_short/2024-10-10_Muon/train_gpt2.py

median runtime for Muon not based on the PyTorch implementation, but based on a custom version that mirrors the implementation details of Muon as closely as possible. The reason for this deviation is that the PyTorch implementation does not provide a version which distributes the optimizer step across ranks, making a fair comparison in this setup infeasible.

Qwen2-0.5B (Table 4). We use the model configuration available on HuggingFace³, but lower the base frequency of RoPE to 10,000 to align the configuration with the initial phase of pre-training as outlined in (Yang et al., 2024a, Section 3.2). We use the 10B split of FineWeb-Edu tokenized with the Qwen2-Tokenizer. We use a grid of $\eta \in \{10^{-3}, 2 \times 10^{-3}, 4 \times 10^{-3}, 8 \times 10^{-3}, 10^{-2}\}$ and tune $\lambda \in \{0, 0.1\}$ for Muon.

Magnitude Optimizer Ablation (Table 5). We adopt the 500M setup and compare three choices for the magnitude optimizer in Algorithm 1 across $\eta \in \{1, 2, 4\} \times 10^{-3}$, all run with $\lambda = 0$: *Fixed* freezes $\mathbf{g}_t = \mathbf{g}_1$ at initialization, exposing the reparameterization without optimizing it; *Signum* replaces Adam with signSGD with momentum, and *Adam* is the default of Algorithm 1.

Licenses. The FineWeb-Edu dataset (Lozhkov et al., 2024) is released under the Open Data Commons Attribution License (ODC-By). plainLM (Ajroldi, 2024) and modded-nanoGPT (Jordan et al., 2024a) are released under the MIT License. Qwen2 (Yang et al., 2024a) is released under the Apache 2.0 License.

³<https://huggingface.co/Qwen/Qwen2-0.5B/blob/main/config.json>